

Upper bounds and anticipative algorithms for the integrated wind farm layout and cable routing problem

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Abstract

To accelerate the green transition it is necessary to decrease the cost of sustainable energy. For offshore wind farms, the costs are largely influenced by the turbine placement and the interconnecting cable network. The two individual problems have been studied intensively in the literature, but since they have contradicting objectives, a unified framework is necessary to avoid suboptimal choices. In this paper we present several upper and lower bounds on the cabling costs to each turbine. By incorporating the bounds into the turbine placement model, it enables finding a good compromise between placement and cable costs.

The bounds on cable costs are further combined into overall upper and lower bounds for the unified layout and cabling problem. To the best of our knowledge these are the first polynomial bounds presented in the literature, and we experimentally show that they are relatively tight.

Furthermore, we present a semi-integrated solution method, where the placement problem is solved with a Variable Neighborhood Search algorithm, using a weighted sum of upper and lower bounds on cabling costs to guide the search. The resulting placement is next improved by a matheuristic to optimize the final cabling layout.

Computational experiments are reported for realistic instances with over 21.000 locations and over 150 turbines. The results show that on average more than 2 million euro can be saved compared to state-of-the-art methods, while computational times are relatively short.

1 Introduction

The Paris Agreement [32], aims to limit the increase in global temperature to well below 2°C. In order to reach this goal it is necessary to have access to affordable, reliable, and sustainable energy. Wind power is an important tool in reaching the sustainability goals. In 2022, wind power accounted for 53.6% of domestic electrical supply in Denmark [1]. To attract even more companies to invest in renewable energy, the costs of producing green energy must be competitive with common, non-sustainable sources.

The life cycle of a wind farm consists of six stages: project development, bidding, detailed design, construction, operation & maintenance, and decommissioning [4]. To make renewable energy more competitive it is important to reduce costs and increase productivity in all these phases. In this

paper we study the placement of turbines and the corresponding cabling. In recent years advanced optimization models have been developed for each of the two parts, but due to the complexity of solving each step, the two optimization problems are nearly always solved sequentially.

Placement is influenced by multiple factors, including the predicted power output at each location based on expected wind patterns. Additionally, soil conditions and sea depth play a crucial role in determining construction costs. Another vital consideration is the *wake effect* — a phenomenon where turbines disrupt wind flow, effectively “stealing” wind from those behind them and reducing overall energy production. In other words, a turbine *shadows* for downwind turbines. Optimizing turbine placement is therefore a balancing act between maximizing power generation and minimizing losses caused by wind interference.

Once the locations are set, the next challenge is to connect the turbines to a *substation*—whether directly or by weaving through other turbines. From the substation, electricity is transmitted onshore using an export cable, which is outside the scope of this paper. Several factors come into play in the cabling process, including cable capacities, costs, restricted areas within the wind farm, and power losses due to resistance.

Naturally, longer cables drive up costs and energy loss—implying that closely spaced turbines are more economical. However, this contradicts turbine placement principles, where spacing them farther apart minimizes the wake effect. These two challenges pull in opposite directions, meaning that an ideal turbine layout minimizing wake might come with steep cabling costs, ultimately undermining the project’s profitability.

Figure 1 compares the solution found by a sequential approach (first layout then cabling) versus an integrated approach for Instance C (Cazzaro and Pisinger [6]). It is seen that the sequential approach leads to turbine placements that make the cabling very expensive, while the integrated approach avoids long cabling connections, leading to an overall saving of approximately 1 million euros.

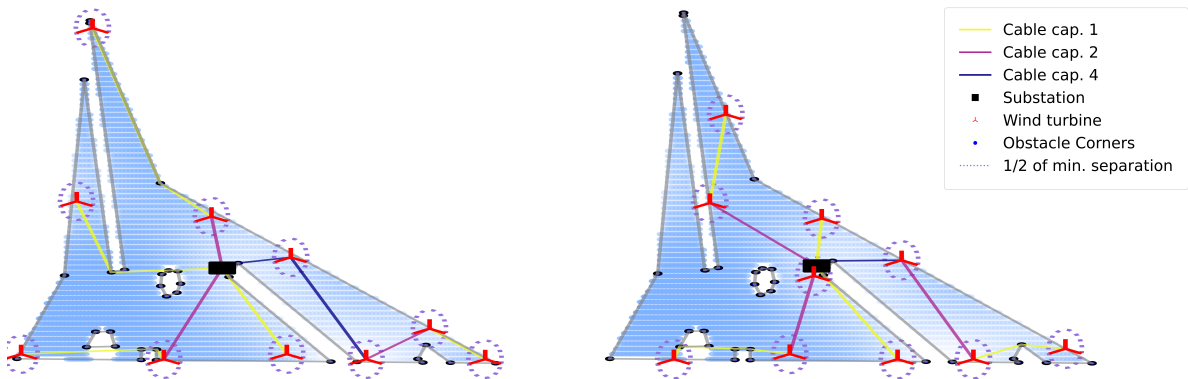


Figure 1: Comparison between sequential and integrated solution approach for Instance C, zone 1. (left) Layout and cabling solved sequentially. (right) Integrated solution of layout and cabling problem. The color of the blue dots represent the construction cost at each position (darker dots are more expensive).

Previous work has addressed the simultaneous optimization of cabling and turbine placement using either branch-and-cut algorithms [16] or metaheuristics [8]. The former is effective only for small instances, while the latter involves high complexity, as it modifies both the cabling and placement at each iteration. In contrast, we demonstrate that by incorporating a simple *estimate* of cabling costs during the turbine placement phase, companies can achieve significant cost savings—on the order of millions of euros—without the need for intricate solution approaches. This method enables the use of existing, scalable, and established techniques for solving the cabling and placement problems separately.

The main **contribution** of this paper is to present an integrated solution algorithm for turbine layout and cabling. As part of this work, we present a series of upper and lower bounds on for the integrated problem. All bounds can be calculated in polynomial time, and we show that they are relatively tight, having an average gap around 5%. The bounds make use of a pre-calculated *estimate* of the cabling costs when placing turbines, allowing us to make use of existing established solution methods for the two separate problems. Computational experiments show that the unified framework is able to solve large-scale instances from the literature in very short time, yielding higher expected profits than those reported in previous studies.

The paper is **organized** as follows. Section 2, gives an overview of previous work on the topic, before separate mathematical models for the two problems are defined in Section 3. In Section 4, upper and lower bounds on the cabling costs are defined and proved, before it is showed how these can be considered when choosing the individual turbine placements. Subsequently, the general solution method is introduced in Section 5, before various implementation details are outlined in Section 6. This includes, for example, strategies for overcoming obstacles. The data utilized for benchmarking is presented in Section 7, leading up to the computational experiments in Section 8. The paper is concluded in Section 9 with a discussion of the results and directions for future work.

This paper builds upon the master thesis of Rustad [31]. Since the completion of the thesis, the solution methods and theorems have been improved and extended.

2 Literature review

This section gives a brief overview of the progress in the *wind farm layout problem* (WFLOP) and *wind farm cable routing problem* (WFCRP), along with recent work in a combined solution approach.

The WFLOP problem consists of placing q wind turbines within a given area, A , while maximizing power production and minimizing installation costs. It was first studied by Kaminsky [25] for a set of turbines on a line. The more realistic two-dimensional version was introduced by Mosetti et al. [27], who proposed a genetic algorithm that allows heuristically solving instances with up to 100 positions.

Chen et al. [11] was among the first to propose methods of solving large-scale instances with up to 5376 available positions. A nested genetic algorithm [30] was employed, in which an outer

genetic algorithm generated turbine layouts, and each layout was subsequently assigned a height configuration by an inner genetic algorithm. Fischetti et al. [17] proposed a matheuristic based on *proximity search* for the WFLOP making it possible to solve instances with up to 20,000 available positions.

Cazzaro and Pisinger [8], used variable neighborhood search (VNS) [22] for the WFLOP, making it possible to solve instances with over 21,000 available positions. The method includes a shake phase that diversifies the search by modifying the current solution using various strategies, such as randomly moving turbines. Moreover, a local search phase is applied, using 1-opt (move one turbine to a better nearby position) and 2-opt (simultaneously move two turbines to the best pair of nearby positions) to improve the solution until no further gains are possible.

It is generally difficult to compare the solution quality of WFLOP algorithms since the individual authors consider different wind farms, and no efficient upper bounds have been available to assess the quality of the solutions.

Having found a good placement of turbines by solving the WFLOP, the *wind farm cabling problem* WFCRP aims to determine the interconnection cabling.

The first mixed integer programming (MIP) model for the WFCRP with multiple cable types was introduced by Fagerfjäll [13]. Ways of avoiding cable crossings was proposed by Bauer and Lysgaard [2], who formulated the WFCRP as a Vehicle Routing Problem (VRP) and solved it using the Clark and Wright savings heuristic. Given the large number of potential cable crossings, and the corresponding constraints, the problem is addressed by iteratively adding only the violated constraints to the model, continuing this process until all constraints are satisfied.

Handling power losses due to resistance in the cables is considered in Fischetti and Pisinger [18]. Using the fact that all turbines have a similar energy production, the authors introduce a preprocessing method to model power losses by replacing each original cable type with a set of sub-cables of varying capacities. These sub-cables incorporate both the installation cost and the expected revenue loss due to power dissipation, which increases with the number of turbines connected.

A number of heuristic algorithms to handle the cable routing, such as simulated annealing, tabu search and VNS, were introduced by Cazzaro et al. [9]. The authors concluded that tabu search and VNS performed the best. In a following paper Cazzaro and Pisinger [7] showed that using a matheuristic approach outperforms the basic heuristics of the previous article.

The *integrated WFLOP and WFCRP* problem (IWFLOP) was first studied by Fagerfjäll [13] using a MIP formulation, but the author concluded that the problem was only solvable for small instances due to computational complexity. Fischetti and Fischetti [16] proposed a branch-and-cut method for the IWFLOP that was tested on instances with up to 100 positions. The first model to run on large-scale instances (over 21,000 positions) was proposed by Cazzaro and Pisinger [8]. The authors outlined a VNS approach to solve the integrated problem. Compared to the integrated model of Fischetti and Fischetti [16], separate models are defined for the WFLOP and WFCRP. In each iteration of the VNS, the solution of the WFLOP is modified, followed by the WFCRP. The

authors concluded that the combined approach showed notable improvements from the sequential approach.

The solution approach presented in this paper for solving the IWFLOP differs significantly from the approaches of Fischetti and Fischetti [16] and Cazzaro and Pisinger [8], by using a semi-integrated method that incorporates upper and lower bounds on the cable costs to each turbine *into* the WFLOP. The goal of the semi-integrated method is to guide the WFLOP solution in a direction that leads to an overall highly profitable solution in the IWFLOP. This approach allows us to leverage existing, scalable, and established techniques to separately solve the WFLOP and WFCRP, while simultaneously avoiding excessive cable costs that would otherwise arise from ignoring cabling considerations during the WFLOP. Overall, the semi-integrated approach leads to a scalable solution method, that is able to find better overall solutions than previous techniques, and allows us to derive polynomial bounds on the solution quality.

3 Formal Model

In this paper we solve two problems, the wind farm layout problem (WFLOP) and the wind farm cable routing problem (WFCRP), using a semi-integrated approach. The goal of the WFLOP is to choose the optimal placement of wind turbines in order to maximize revenue from power production. Given a set of turbine locations, the aim of the WFCRP is to find the cheapest way of passing the electricity from the turbines to the substation. This section provides a formal definition of the two problems, beginning with the WFLOP. In section 4, we demonstrate how to incorporate an estimate of the cable cost *into* the objective function of the WFLOP, thereby developing a semi-integrated approach.

3.1 Wind Farm Layout Problem (WFLOP)

The WFLOP consists of placing a fixed number of turbines within a given area A , while maximizing power production and minimizing installation costs. Within A it is common to have restricted areas, denoted *obstacles*, where turbines cannot be placed. To avoid damage due to turbulence from other turbines, a minimum separation distance δ between turbines must be upheld. Furthermore, the area A can be treated as either a continuous space or discretized into N available positions. In this paper, the latter approach is considered.

It is assumed that the following input is available: A set N of *available turbine positions*, the *construction costs* B_i at position $i \in N$, the requested number q of turbines to place, and a *minimum distance* δ between turbines.

Furthermore, the input also covers a *power and wake matrix*, P , of size $|N| \times |N|$. The diagonal elements $P_{i,i} \geq 0$ represents the power production at position i in free wind, namely with no other turbines nearby. An element $P_{i,j} \leq 0$, where $i \neq j$, corresponds to the loss of power at position j due to wake from a wind turbine placed at position i . We suppose a linearly additive wake effect. That is, by summing $P_{i,j}$ over all positions i where a turbine is located, we can quantify the total

wake effect exerted on the turbine placed at j . In order to compare the values of the power matrix, P , to the costs in B , the values in P are converted from units of power (MW) to units of revenue ($M\text{€}$). This is done using an *energy price factor* (EPF) that will be defined in subsection 7.5.

The minimum distance constraint, saying that $\text{dist}(i, j) \geq \delta$, can be incorporated into the objective function by penalizing closely placed turbines. This is done by defining a new matrix R where a penalty of $-M$ is introduced to $P_{i,j}$, if turbines i, j do not satisfy the minimum distance δ . In addition, the construction cost can be included in R by subtracting B_i from the diagonal elements. The definition of R hence becomes:

$$R_{i,j} = \begin{cases} P_{i,j} - B_i, & \text{if } i = j, \\ -M, & \text{if } \text{dist}(i, j) < \delta \wedge i \neq j, \\ P_{i,j}, & \text{otherwise.} \end{cases} \quad (1)$$

Introducing binary variables x_i to indicate if a turbine is placed at position i we can formulate the WFLOP as the following quadratic problem denoted (QWFP):

$$\max \quad z_{\text{layout}} = \sum_{i \in N} \sum_{j \in N} \Pi_{i,j} x_i x_j, \quad (2)$$

$$\sum_{i \in N} x_i = q, \quad (3)$$

$$x_i \in \mathbb{B}, \quad \forall i \in N \quad (4)$$

where we use $\Pi = R$ given by (1). Later in the paper we will use alternative formulations of Π .

The problem is now roughly reduced to the *p-dispersion sum problem* (PDSP) [29] of size $O(|N|^2)$, which is defined as follows: given a set of n predefined locations, the goal is to place p facilities to maximize the sum of the distances between them. In our case, there are $|N|$ predefined available turbine positions and q turbines to place. The only difference is that in the PDSP, we have $\Pi_{i,i} = 0$. This can be accounted for by adding $\Pi_{i,i}/(2q)$ to $\Pi_{i,j}$ and $\Pi_{j,i}$ for all $j \in N$ such that $j \neq i$, before setting $\Pi_{i,i} = 0$. The problem itself, including the objective value, would stay unchanged.

By formulating the WFLOP as a PDSP, we can make use of a variety of upper bounds and approximation algorithms for PDSP, described in the survey by Pisinger [29]. This will further be used to derive efficient bounds for the integrated layout and cabling problem.

3.2 Wind Farm Cable Routing Problem (WFCRP)

The WFCRP aims to efficiently connect all turbines to a substation, either directly or through other turbines, while avoiding restricted areas. From the substation, an export cable transmits the energy to customers on-shore. There are several possible topologies for turbine connections, each representing a different method of linking them. A *stringed* topology allows for only one cable to enter and exit at each turbine, while a *branched* topology allows multiple cables to enter and one to

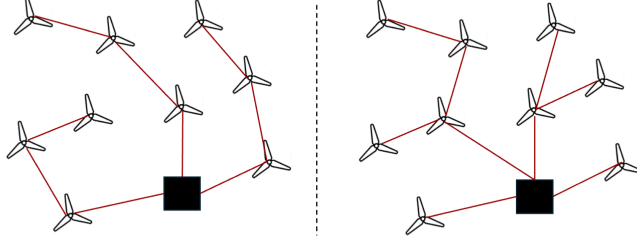


Figure 2: An example of a string topology (left) and a branched topology (right). The black square represents the substation.

exit. An example of the two topologies is shown in Figure 2. Moreover, a k -constrained topology accepts at most k cables entering the substation. With the positions of the turbines determined by the WFLOP, the model of the WFCRP introduced in this paper assumes a stringed topology, without any constraint on the number of cables entering the substation. Other topologies are, however, easy to implement.

To formally describe the WFCRP, let T be the set of all turbines to be connected, and node s be the substation. Furthermore, let C be the set of all cable types, where we for each type $i \in C$ have the cost per meter a_i , and capacity c_i (measured in number of supported turbines). Moreover, define $V = T \cup \{s\}$, i.e. the set of all vertices. E_X is the set of arcs that intersect. Finally, we can in polynomial time calculate the shortest distance $D_{i,j}$ between all pairs of vertices, avoiding obstacles.

We have two sets of decision variables. The binary variable $x_{i,j,c}$ for $i, j \in V$ and $c \in C$ indicates whether or not a cable of type c is placed between vertex i and j . Further, the integer flow variable $f_{i,j}$ for $i, j \in V$ denotes the transmitted power (measured in number of turbines) by the cable between vertices i and j .

$$\min \quad z_{cable} = \sum_{i \in V} \sum_{j \in V} \sum_{c \in C} D_{i,j} a_c x_{i,j,c} \quad (5)$$

$$s.t. \quad \sum_{j \in V} \sum_{c \in C} x_{i,j,c} = 1 \quad \forall i \in T \quad (6)$$

$$\sum_{j \in V} \sum_{c \in C} x_{s,j,c} = 0 \quad (7)$$

$$\sum_{j \in V} \sum_{c \in C} x_{j,i,c} \leq 1 \quad \forall i \in T \quad (8)$$

$$x_{i,i,c} = 0 \quad \forall i \in V, c \in C \quad (9)$$

$$f_{i,j} \leq \sum_{c \in C} c_c \cdot x_{i,j,c} \quad \forall i, j \in V \quad (10)$$

$$\sum_{j \in V} f_{i,j} = \sum_{j \in V} f_{j,i} + 1 \quad \forall i \in T \quad (11)$$

$$\sum_{j \in V} f_{j,s} = |T| \quad (12)$$

$$\sum_{c \in C} x_{i,j,c} + \sum_{c \in C} x_{k,l,c} \leq 1 \quad \forall ((i,j), (k,l)) \in E_X \quad (13)$$

$$x_{i,j,c} \in \mathbb{B} \quad \forall i, j \in V, i \in C \quad (14)$$

$$f_{i,j} \in \mathbb{N} \quad \forall i, j \in V \quad (15)$$

The MIP model shown in (5) through (15) is a modified version of the formulation introduced by Cazzaro and Pisinger [7]. The objective value (5), minimizes the total cable cost z_{cable} . Constraints (6) and (7) ensure that one edge leaves each turbine and that there are no cables leaving the substation. Constraints (8) ensure that only one ingoing cable at each turbine, defining a stringed topology. No self looping cables are established through (9), and (10) says that the number of turbines supported by a cable cannot exceed the cable's capacity. Moreover, (11) ensures that the flow leaving a turbine equals the incoming flow, plus one. Constraint (12) states that the flow into the substation must equal the number of turbines. Further, constraints (13) ensure that we have no crossing edges. Finally, (14) and (15) defines the domain of the decision variables.

The model contains $O(|V|^2 + |V||C|)$ constraints without the non-crossing constraints, and $O(|V|^4 + |V||C|)$ constraints when they are included. We have $O(|V|^2|C|)$ binary variables and $O(|V|^2)$ integer variables.

3.3 Integrated Wind Farm Layout Problem (IWFLOP)

The integrated wind farm layout problem aims at maximizing the overall objective value $z_{int} = z_{layout} - z_{cable}$ where z_{layout} is a solution to WFLOP, and z_{cable} is a solution to the corresponding WFCRP. Integrated models, however, tend to become very large since in the WFCRP we potentially can have a turbine at every position N meaning that we have $O(|N|^2)$ possible edges and $O(|N|^4)$

non-crossing constraints. In sequential approaches, first solving WFLOP and then WRCRP, the positions of the turbines are given as input to the WRCRP, making the latter problem much more tractable.

4 Bounds on cabling costs

Due to the complexity of the WFLOP and the WFCRP, most solution methods solve the problems sequentially. However, Cazzaro et al. [10] showed that a sequential approach may lead to very bad solutions, hence from an economical perspective an integrated approach is preferred. In this section we introduce some upper and lower bounds on the cable cost to each turbine location, and we conclude the section by showing how this knowledge can be incorporated into the WFLOP problem, enabling a semi-integrated approach. We will call this an anticipative solution approach.

4.1 Upper and lower bounds on cabling costs

The subsection begins by introducing and proving an initial upper and lower bound for the WFCRP, given the turbine positions. This is done within the first two propositions. Moreover, we propose and validate four distinct methods for lifting the lower bound. All proofs are found in the appendix.

All definitions and assumptions from section 3 hold, with the addition of three assumptions:

- (i) The cables are sorted according to cost, i.e. $a_1 < a_2 < \dots < a_n$ and capacity $c_1 < c_2 < \dots < c_n$. If this is not the case, dominated cable types can be removed.
- (ii) Cable types satisfy economy of scale, meaning that cables with lower costs support fewer turbines and thus have a higher average cost per turbine. Mathematically, we have $a_1/c_1 \geq a_2/c_2 \geq \dots \geq a_n/c_n$. With this assumption, $n \in C$ is the most cost efficient cable type.
- (iii) A turbine $i \in T$ is located d_i units away from the substation.

Some of the bounds hold for a branched topology, while we also present specialized bounds for string topology. Since the string topology is a subset of the branched topology, bounds for branched topology also hold for string topology.

4.1.1 Upper bound

We start by defining an upper bound. The idea is as follows: individually connecting all turbines to the substation using the cheapest cable is a feasible solution, and hence, the associated objective value is an upper bound.

Proposition 1. *Assume there is no limit to the number of cables connected to the substation. Then*

$$\mathcal{U} = \sum_{i \in T} d_i a_1, \tag{16}$$

is an upper bound on the total cable cost.

Proof. See Appendix. □

4.1.2 Initial lower bound

We proceed by proposing an initial lower bound. In the best-case scenario each turbine is connected to the substation using the most cost efficient cable, $n \in C$ and sharing the cost with the maximum number of turbines, c_n , as given by the capacity of the cable. This is the principle underlying the lower bound.

Proposition 2.

$$\mathcal{L}_1 = \sum_{i \in T} d_i \frac{a_n}{c_n}, \quad (17)$$

is a lower bound on the total cable cost.

Proof. See Appendix. □

Computing the upper and lower bounds can be done in $O(q)$ time, given that d_i is known for all $i \in T$.

4.1.3 Lifting the lower bound - leaf nodes

Now that we have provided an initial upper and lower bound, we want to shrink the gap. The initial lower bound, $\mathcal{L}_1$, considers the ideal case where the cost of all cables are shared among multiple turbines. Unfortunately, besides in some very special cases, this cannot be the case. Consider for instance the leaf node on a string, i.e. farthest from the substation. It must be connected to either the substation or another turbine. Being the leaf turbine on the string, we know that the outgoing cable supports only one turbine. Hence, any optimal solution would use the cheapest cable, c_1 . Moreover, the cable must have a minimum length δ . This allows us to add $c_1\delta$ to the initial lower bound $\mathcal{L}_1$. This can be applied to all leaf node turbines in a wind farm. Moreover, the idea can be extended beyond the first turbine on the string. However, when adding to the initial lower bound, we must account for the fact that parts of the cabling cost have already been included in $\mathcal{L}_1$, and therefore must be subtracted to avoid double-counting. This leads us to Proposition 3.

Proposition 3. *Assume that the cabling should satisfy a branched topology. Moreover, assume that the distance to the substation exceeds $g \cdot \delta$ for all but m turbines, where $g \in \mathbb{N}$ and $g \leq c_1$. Define $G = \lceil \frac{g-m}{c_n} \rceil$. Then*

$$\mathcal{L}_2 = G\delta \left(ga_1 - \frac{a_n}{c_n} \sum_{i=1}^g i \right) + \sum_{i \in T} d_i \frac{a_n}{c_n}, \quad (18)$$

is a lower bound on the total cable cost.

Proof. See Appendix. □

Calculating $\mathcal{L}_2$ has a running time of $O(g + q)$, given d_i for all $i \in T$.

4.1.4 Lifting the lower bound - imperfect cabling

The previous subsection presented one approach to lifting $\mathcal{L}_1$ for a branched topologies. In this subsection, we introduce an alternative method to increase the lower bound.

As mentioned, the ideal lower bound, $\mathcal{L}_1$, is rarely the actual cable cost, due to "imperfect" cabling. As an example, we may in a concrete solution have a cable of capacity 4 supporting only 1 turbine. In fact, if the cheapest cable has a capacity of 4 turbines, this will have to be the case for all leaf-node turbines, i.e. the turbines without any input cables. In this case, we can say that 1 turbine is paying for something that could have been paid for by 4 turbines.

Before generalizing, we proceed by showing an example of how one can calculate the loss due to "imperfect" cabling around a turbine.

Example 1. *In this example, we have two cable types: Cable type A, which supports 4 turbines at a cost of 5 per unit. Cable type B, which supports 7 turbines at a cost of 8 per unit. We consider a string topology, so we only have 7 different cases for ingoing and outgoing connections.*

In Table 1 we show how to calculate the cable loss due to imperfect cabling at each turbine. The first columns shows the ingoing and outgoing cable cost for various scenarios, in addition to the lower bound, i.e. the most cost-efficient cabling where the cost of the most expensive cable is shared among the maximum number of supported turbines.

Number of turbines in	Number of turbines out	Cheapest cable type in	Cheapest cable type out	Lower bound in	Lower bound out	Loss in	Loss out	Loss around turbine
0	1	0	A	0	$1 \cdot \frac{8}{7}$	0	$5 - \frac{1 \cdot 8}{7}$	$\frac{1}{2}\delta(0 + 5 - \frac{1 \cdot 8}{7})$
1	2	A	A	$1 \cdot \frac{8}{7}$	$2 \cdot \frac{8}{7}$	$5 - \frac{1 \cdot 8}{7}$	$5 - \frac{2 \cdot 8}{7}$	$\frac{1}{2}\delta(5 - \frac{1 \cdot 8}{7} + 5 - \frac{2 \cdot 8}{7})$
2	3	A	A	$2 \cdot \frac{8}{7}$	$3 \cdot \frac{8}{7}$	$5 - \frac{2 \cdot 8}{7}$	$5 - \frac{3 \cdot 8}{7}$	$\frac{1}{2}\delta(5 - \frac{2 \cdot 8}{7} + 5 - \frac{3 \cdot 8}{7})$
3	4	A	A	$3 \cdot \frac{8}{7}$	$4 \cdot \frac{8}{7}$	$5 - \frac{3 \cdot 8}{7}$	$5 - \frac{4 \cdot 8}{7}$	$\frac{1}{2}\delta(5 - \frac{3 \cdot 8}{7} + 5 - \frac{4 \cdot 8}{7})$
4	5	A	B	$4 \cdot \frac{8}{7}$	$5 \cdot \frac{8}{7}$	$5 - \frac{4 \cdot 8}{7}$	$8 - \frac{5 \cdot 8}{7}$	$\frac{1}{2}\delta(5 - \frac{4 \cdot 8}{7} + 8 - \frac{5 \cdot 8}{7})$
5	6	B	B	$5 \cdot \frac{8}{7}$	$6 \cdot \frac{8}{7}$	$8 - \frac{5 \cdot 8}{7}$	$8 - \frac{6 \cdot 8}{7}$	$\frac{1}{2}\delta(8 - \frac{5 \cdot 8}{7} + 8 - \frac{6 \cdot 8}{7})$
6	7	B	B	$6 \cdot \frac{8}{7}$	$7 \cdot \frac{8}{7}$	$8 - \frac{6 \cdot 8}{7}$	$8 - \frac{7 \cdot 8}{7}$	$\frac{1}{2}\delta(8 - \frac{6 \cdot 8}{7} + 8 - \frac{7 \cdot 8}{7})$

Table 1: Calculation of the loss due to "imperfect" cabling around a turbine for each of the possible number of in- and outgoing connections.

Using the cheapest cable cost, as well as the calculated lower bound, both going in and out, we can estimate the loss around the turbine. As all cables between turbines have a length of at least δ , we consider the loss around a turbine to be due to a cable distance of $\frac{1}{2}\delta$ before the turbine and $\frac{1}{2}\delta$ after the turbine. In this way, no loss estimates overlap between turbines.

The cable loss going in and out of a turbine is found by taking the cheapest cable cost for a specific scenario and subtracting the lower bound, i.e. the scenario where we use the most cost-efficient cable and there is no loss due to "imperfect" cabling. Finally, the loss around a turbine is found by multiplying the loss going in and out by $\frac{1}{2}\delta$, before adding them together. In other words,

the loss around a turbine is defined within the radius $\frac{1}{2}\delta$ of the turbine.

Since every turbine must follow one of the above cases, we can take the minimum loss in the table, and add it to all nodes. In this way we can lift the lower bound.

We now generalize the loss calculation at a turbine, starting with a *stringed topology*, before generalizing to a branched topology.

Lemma 1. *Remember that c_n denotes the capacity of the cable with the highest capacity. Let i be any integer $1 \leq i \leq c_n$. Assume that cable type j is the cheapest cable supporting i turbines and cable type h is the cheapest cable type supporting $i-1$ turbines (note that h may equal j). Moreover, assume $a_0 = c_0 = 0$. Define*

$$\ell_i^s = \frac{1}{2}\delta \left(a_h + a_j - ((i-1) + i) \frac{a_n}{c_n} \right).$$

Then, given a stringed topology, ℓ_i^s is the cable loss for a turbine with an input and output cable supporting $i-1$ and i turbines, respectively.

We proceed by generalizing to a *branched topology*, which requires an additional lemma.

Lemma 2. *For $0 \leq j \leq c_n$, define*

$$g_0 = 0, \quad g_j = \min_{i \in C} \left(a_i + g_{\max(j-c_i, 0)} \right).$$

Then g_j denotes the minimum cable cost of supporting j turbines, when allowing for the use of multiple cables between two turbines.

Lemma 3. *Let $1 \leq i \leq c_n$. Let $h \in C$ denote the cheapest cable that can support i turbines. Define*

$$\ell_i^b = \frac{1}{2}\delta \left(g_{i-1} + a_h - ((i-1) + i) \frac{a_n}{c_n} \right).$$

Then ℓ_i^b is the minimum cable loss for a turbine with incoming cables supporting $i-1$ turbines in total and an outgoing cable supporting i turbines.

With all lemmas introduced, we are ready to lift the lower bound.

Proposition 4. *Set*

$$\ell_{\min} = \begin{cases} \min_{1 \leq i \leq c_n} \ell_i^s & \text{if stringed topology} \\ \min_{1 \leq i \leq c_n} \ell_i^b & \text{if branched topology} \end{cases} \quad (19)$$

Assume there are at most m turbines placed within a radius $\frac{1}{2}\delta$ of the substation. Then

$$\mathcal{L}_3 = (q - m)\ell_{\min} + \sum_{i \in T} d_i \frac{a_n}{c_n}, \quad (20)$$

is a lower bound on the total cable cost.

Proof. See Appendix. □

Calculating the lower bound is done in $O(c_n + q)$ time for a stringed topology and $O(c_n^2 + q)$ for a branched topology, assuming d_i is known for all q turbines.

4.1.5 Lifting the lower bound - minimum distance

Next, we introduce a lower bound based on the minimum distance between turbines. All q turbines must be connected to a turbine or the substation. The cable must have a minimum length δ and it cannot be cheaper than a_1 .

Proposition 5. *Assume that the cabling should satisfy a branched topology and that there are at most m turbines within a radius of distance δ from the substation. Then*

$$\mathcal{L}_4 = (q - m)\delta a_1 \tag{21}$$

is a lower bound on the cable cost.

Proof. See Appendix. □

If the number m of turbines within a radius of distance δ from the substation is known, the lower bound can be computed in $O(1)$ time. Otherwise, it is done in $O(q)$ time, assuming d_i is known for all $i \in T$.

4.1.6 Lifting the lower bound - max bound

If we take the maximum of the four bounds from Proposition 2–5 we obviously get an even tighter lower bound.

Corollary 1. *Assume that the cabling should satisfy a string topology and that there are at most m turbines within a radius of distance δ from the substation. Then*

$$\mathcal{L}_5 = \max\{\mathcal{L}_1, \mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_4\} \tag{22}$$

is a lower bound on the total cable costs.

4.2 Application to the Integrated Wind Farm Layout Problem

With the theorems established, the bounds from Propositions 1 and 2 can now be applied to the Integrated Wind Farm Layout Problem. These propositions provide insight into each turbine's contribution to the upper and lower bounds given the distance to the substation. Incorporating this into the WFLOP helps ensure that turbines are not placed in areas with expected high cable costs, i.e. far away from the substation. For instance, if the cable cost is significantly higher than

the marginal increase in power production from increasing turbine separation, it is preferable to keep the turbines closer together.

As established in the proof of Proposition 1 and 2, the “contribution” of turbine $i \in T$ to the lower and upper bound is $l_i = d_i \frac{a_n}{c_n}$ and $u_i = d_i a_1$, respectively. Hence, we can estimate the cabling costs for turbine i in the WFLOP by choosing any value between l_i and u_i . For a given $\alpha \in [0, 1]$, we define a vector L^α of *estimated* cable costs where

$$L_i^\alpha = l_i + \alpha(u_i - l_i) \quad \forall i \in T. \quad (23)$$

Recall from subsection 3.1 that the matrix R is a modified version of the power matrix, obtained by subtracting construction costs from the diagonal elements and adding penalties to the off-diagonal elements associated with closely spaced positions. The cable estimate for each available turbine position is incorporated into the matrix Π by subtracting L^α from the diagonal of R as follows:

$$\Pi_{i,i} = R_{i,i} - L_i^\alpha \quad \forall i \in T. \quad (24)$$

4.3 Bounds on the Integrated Problem

The upper and lower bounds introduced in this section can be combined with the WFLOP to define bounds on the integrated problem IWFLOP. Specifically, given any feasible solution to the WFLOP, the WFCRP upper bound, U , can be subtracted from the WFLOP objective value, z_{layout} , to derive a lower bound on the integrated objective. In the following section, 5, we describe how such a feasible WFLOP solution can be obtained.

Proposition 6. *The QWFP model given by (2)–(4) provides a lower bound on the integrated problem IWFLOP, when using $\alpha = 1$ in the definition of $\Pi_{i,i}$ in (24). Call this $\mathcal{L}^{\text{int}}$.*

Proof. See Appendix. □

An upper bound on the integrated problem IWFLOP can be obtained by first redefining Π using a lower bound estimate of the cabling cost (i.e., setting $\alpha = 0$), and then calculating an upper bound on the modified WFLOP. Since the QWFP given by (2)–(4) is a PDSP we can use any of the polynomial bounds from Pisinger [29]. In this paper, we use the bound $\mathcal{U}_1^{\text{pdsp}}$ from [29], which can be calculated in $O(|N|^2)$ time.

Proposition 7. *The QWFP model provides an upper bound on the integrated problem IWFLOP, when using $\alpha = 0$ in the definition of $\Pi_{i,i}$ in (24). Call this $\mathcal{U}_1^{\text{int}}$.*

Proof. See Appendix. □

The upper bound on the Integrated problem IWFLOP defined in Proposition 7 can be tightened (lowered) by redefining the matrix Π using an idea from Proposition 3. In particular, the matrix Π in (24) can be further redefined as follows:

$$\Pi_{i,i} = \begin{cases} R_{i,i} - \frac{\delta}{c_n} \left(a_1 - \frac{a_n}{c_n} \right) - L_i^\alpha & \forall i \in T, d_i \geq \delta, \\ R_{i,i} - L_i^\alpha & \text{otherwise.} \end{cases} \quad (25)$$

Proposition 8. *The QWFP model provides an upper bound on the Integrated problem IWFLOP, when using $\alpha = 0$ in the definition of $\Pi_{i,i}$ in (25). Call this $\mathcal{U}_2^{int}$.*

Proof. See Appendix. □

Similarly, the upper bound on the integrated problem can be lowered using ideas from Proposition 4. In this case, we redefine matrix Π in (24) as follows:

$$\Pi_{i,i} = \begin{cases} R_{i,i} - \ell_{min} - L_i^\alpha & \forall i \in T, d_i \geq \frac{1}{2}\delta, \\ R_{i,i} - L_i^\alpha & \text{otherwise.} \end{cases} \quad (26)$$

Proposition 9. *The QWFP model given by (2)–(4) provides an upper bound on the integrated problem IWFLOP, when using $\alpha = 0$ in the definition of $\Pi_{i,i}$ in (26). Call this $\mathcal{U}_3^{int}$.*

Proof. See Appendix. □

If we take the minimum of the three bounds from Proposition 7–9 we obviously get an even tighter upper bound.

Corollary 2.

$$\mathcal{U}_4^{int} = \min\{\mathcal{U}_1^{int}, \mathcal{U}_2^{int}, \mathcal{U}_3^{int}\} \quad (27)$$

is an upper bound on the integrated problem.

Note, a lower bound on the PDSP defined in Proposition 6 can be calculated in $O(n^3)$ time by adopting the greedy heuristic used by Pisinger [29]. Moreover, notice that instead of solving QWFP directly in bounds $(\mathcal{U}_1^{int}, \mathcal{U}_2^{int}, \mathcal{U}_3^{int})$ we can use a polynomial upper bound on the PDSP [29]. In this way we get a family of polynomial bounds for the IWFLOP.

5 Anticipative Algorithm

Now that the formal models have been defined in section 3, and subsection 4.2 showed how to incorporate estimates of the WFCRP solution *into* the WFLOP problem, the solution method for the two problems is introduced. We begin with the WFLOP.

5.1 Wind Farm Layout Problem (WFLOP)

The WFLOP is solved using a metaheuristic approach based on *variable neighborhood search* (VNS) [22]. The main idea of VNS is that multiple neighborhoods are defined, and the algorithm systematically switches between them in each iteration. This allows us to search different solution spaces, increasing the ability to escape local minima. A detailed explanation of VNS will be provided in subsection 5.1.2; however, the first step of the algorithm concerns generating an initial solution.

5.1.1 Wind turbine placement - initial greedy solution

An initial solution to the WFLOP is found by taking advantage of the similarities between our problem and the PDSP, as described in subsection 3.1. The PDSP is NP-hard, so we adopt the $O(n^3)$ greedy heuristic used by Pisinger [29] to the WFLOP. The heuristic begins by placing a turbine at each available position, before removing the “worst” turbine, one at a time. Turbines are removed until we reach the desired number of turbines, q . The worst turbine is considered to be the turbine that contributes the minimum amount to the objective value. For large problems, a number of techniques can be used to speed up the algorithm: Instead of removing the worst turbine in each iteration, one can remove the m worst turbines. Alternatively, using a GRASP framework [14], one can randomly choose and remove m out of a restricted candidate list $\mathcal{R}$ of the worst turbines, where $m \leq |\mathcal{R}|$.

5.1.2 Improvement heuristic

The PDSP heuristic provides a good initial solution in polynomial time. In order to further improve the solution we use a VNS metaheuristic [22], inspired by the work of Cazzaro and Pisinger [8]. We begin by generating an initial solution, as described in the previous subsection. Then, each iteration consists of two main elements: making a random change to the solution (shake up), and searching for new solutions that may lead to improvements (local search). The first step, i.e. the random change, is done by choosing k turbines and moving them to a random, feasible location within the m closest positions. The value k increases in each iteration if no improvement is found and we have not reached a defined maximum value of k . Otherwise, it is reset back to 1. This is repeated until the time limit is reached.

The local search is implemented using a first-improvement 1-opt method. Assume each turbine is associated with a list of the m closest positions. For each turbine in the solution, considered consecutively, all movements within their respective list of m closest positions are tested. Whenever an improvement is found, the turbine is moved to this location. To reduce computational complexity, only the relative change in objective value is calculated in each move.

5.2 Cabling Problem

After a solution to the WFLOP is determined, we want to find a solution to the the MIP model described in subsection 3.2. However, the model may be time consuming to solve for large instances. Therefore, we apply a matheuristic to solve the cabling problem efficiently, based on the techniques in Cazzaro et al. [7]. At a high level, turbines are placed into subgroups that will form a branch from the substation, and the MIP model is solved separately for each subgroup. The final solution is then constructed by combining the results from all groups.

5.2.1 Grouping turbines

Dividing turbines into groups is a key aspect of the cabling math heuristic. Conceptually, we aim to group turbines based on their direction as seen from the substation.

More specifically, the grouping is based on a sweep algorithm [20] adapted to the WFCRP with obstacles as described by Cazzaro et al. [9]. First, the shortest path between all individual turbines and the substation is determined. Then, the turbines are ordered according to the angle of the first edge on their respective shortest path. If multiple turbines share the first edge, then the second edge is considered, etc. The turbines are then grouped based on their ordering, such that a group consists of turbines with a similar start angle. An illustration of the algorithm is shown in Figure 3.

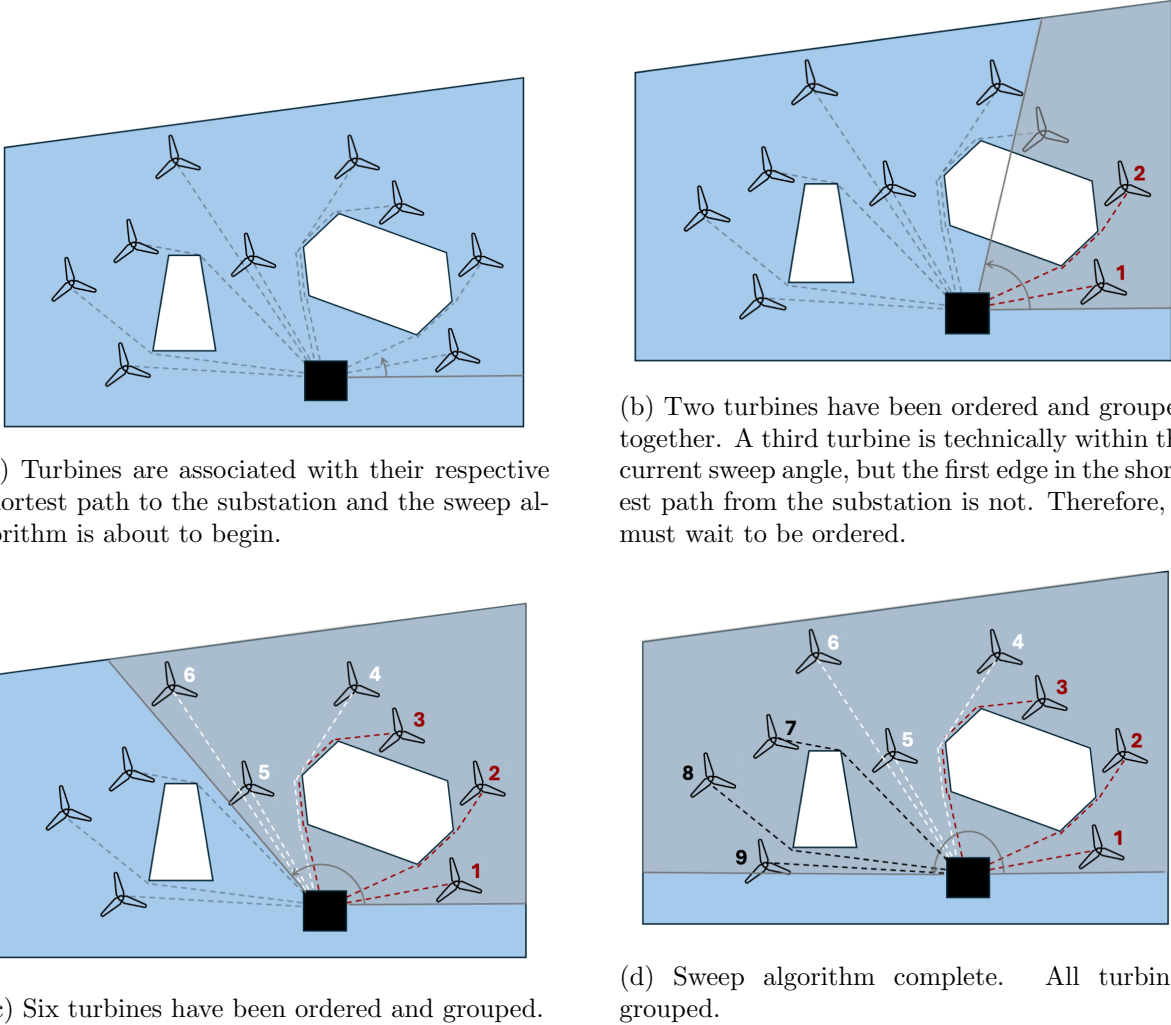


Figure 3: Illustration of the sweep algorithm for grouping turbines that later will form a branch of the cabling. Colors (red, white, black) indicate the individual groups.

5.2.2 Matheuristic

After the turbines have been grouped, the MIP-model (5) – (15) is solved for each group to form the overall solution. Since we cannot be sure that the initial grouping of the turbines is optimal, we use a matheuristic [15] to improve the cabling solution. In this algorithm we use metaheuristics (based on local search and perturbation) to form the groups and MIP solver to solve the individual WFCRP problems for each group.

Before providing the full algorithm, two of the main tools utilized are explained, namely the perturbation method and local search. The perturbation method involves randomly modifying the start angle of the sweep algorithm. Next, some randomness is introduced to the size of the groups, such that group g will have a target size σ_g that is randomly chosen in the interval $[\mu - \nu, \mu + \nu]$, where μ and ν are given parameters. All remaining turbines will be placed in the last group.

The turbine ordering, calculated in the sweep algorithm, is utilized in the local search. The local search consists of moving a small number of turbines to neighboring groups and resolving the MIP model for the modified groups. A first-improvement approach is used, i.e. the solution is modified as soon as an improvement is found. This method is run until no improvements are found. Note, we keep a record of all groups modified without achieving an improved objective value to prevent redundant calculations.

With the perturbation method and local search explained, the matheuristic can be introduced. First, an initial solution is found by applying the sweep algorithm, before a local search is run. In each iteration, we perturbate the groups, followed by a local search. We keep track of the solution with the best objective value and update it whenever an improved solution is encountered. The metaheuristic part of the matheuristic is essentially an *Iterative Local Search* (ILS) [26].

Algorithm 1 gives a high-level overview of the semi-integrated approach, demonstrating the connection between all components presented in this section.

Algorithm 1 Semi-integrated Method

```

1: procedure SEMIINTEGRATED( $T_{\max}$ )
2:   # Layout
3:    $x \leftarrow \text{GreedyInit}()$  ▷ Initial WFLOP solution, Section 5.1.1
4:    $(x, z_{\text{layout}}) \leftarrow \text{ImprovementHeuristic}(x, \frac{1}{2}T_{\max})$  ▷ Improvement heuristic, Section 5.1.2
5:   # Cabling
6:   while time  $\leq T_{\max}$  do
7:      $G \leftarrow \text{GroupTurbines}(x)$  ▷ Sweep / perturbation / local search
8:     for  $g \in G$  do  $(\text{cabling}^g, z_{\text{cable}}^g) \leftarrow \text{MIPSolver}(g)$  ▷ Solve model (5)–(15) for each group
9:      $z_{\text{cable}} = \sum_{g \in G} z_{\text{cable}}^g$ 
10:    if  $z_{\text{cable}} < z_{\text{cable}}^{\text{best}}$  then  $z_{\text{cable}}^{\text{best}} = z_{\text{cable}}$  ▷ Save the best cabling encountered
11:  end while
12:   $z_{\text{int}} \leftarrow z_{\text{layout}} - z_{\text{cable}}^{\text{best}}$ 
13:  return  $z_{\text{int}}$ ,  $x$  and  $\text{cabling}$ 
14: end procedure

```

6 Implementation Details

The previous section provided an overview of the general solution method. This section delves into additional key considerations, including bypassing obstacles and avoiding cable crossings.

6.1 Obstacles

Obstacles are areas where neither turbines nor cables can be installed. Examples of obstacles include designated marine routes, fishing zones, protected nature reserves, and unstable soil. Obstacles, defined as convex polygons, are given as input.

A key aspect of our semi-integrated approach is to associate each available turbine position with an estimated cable cost. This requires knowledge on the shortest path between the substation and each individual turbine, taking into account obstacles. To find this distance we first construct a *visibility graph*: First we define an extended set of nodes V' defined as the union of turbine positions and nodes in the obstacle polygons. All pairs of nodes are connected with an edge, if they do not cross an obstacle. Next, we solve a shortest path problem from the base station to all turbine positions using the resulting graph.

Given b obstacle corners and n available turbine positions, the total running time is $O(b^3 + nb^2)$.

6.2 Cable Crossing

Although it is technically possible to install two intersecting cables, it is highly discouraged due to the substantial increase in costs and maintenance requirements. This is the reasoning behind the cable crossing constraint (13) in the MIP model described in subsection 3.2.

Unfortunately, with $|V|$ vertices and a magnitude of $|V|^2$ edges, this can potentially lead to a number of constraints on the order of $|V|^4$. To avoid the large number of constraints, the MIP model is first run without the constraint in (13). Once the optimizer is done, a check for crossing edges is run. For all intersecting edges in the solution, a constraint prohibiting the intersection is added to the MIP model before the model is rerun.

Dealing with cable crossings is done differently in the cabling matheuristic described in subsection 5.2.2. Although the MIP model avoids intersections within a group of turbines, it does not guarantee no intersections between groups. This is handled by recognizing that the optimal cabling of each group is determined sequentially. Hence, by locking the cables of the previous groups and avoiding all edges that intersect with placed cables, we avoid cable crossings.

7 Data

This section provides an overview of the input data utilized in the project. The greater part stems from the synthetic instances introduced by Cazzaro et al. [6]. They present a set of 10 synthetic site layouts, in addition to a collection of wind and turbine data collected from other sources. The first three subsections, 7.1, 7.2 and 7.3, describe the publicly available data [6]. In subsection 7.4

we describe how the data is used to produce the power matrix, described in subsection 3.1, as well as calculating the wake effect. Subsection 7.5 specifies the conversion from energy production to monetary value. Finally, 7.6 defines the types and specifications of the cables.

7.1 Layout

Cazzaro et al. [6] created a set of 10 realistic layouts, each consisting of 1 to 3 zones. For each site, windmills can be placed in any location within the associated zones. The available positions within the zones are discretized and provided by the authors, along with the depth and foundation cost (in Euros) of mounting a turbine at each respective position. Additionally, the data also includes a file outlining the boundaries of each zone, which is particularly useful when ensuring that cables do not cross zone limits, including obstacles. Further details on how to avoid this can be found in subsection 6.1. A summary of the sites is shown in Table 2.

Synthetic Instance	Available positions	Zones	Number of Turbines to Place
Instance A	3238	2	13, 7
Instance B	6989	1	49
Instance C	7098	2	30, 15
Instance D	10443	1	85
Instance E	11518	3	3, 47, 18
Instance F	11548	2	66, 13
Instance G	14637	1	70
Instance H	19498	2	79, 15
Instance I	20247	1	156
Instance J	21709	3	68, 37, 12

Table 2: Summary of the 10 synthetic sites provided by Cazzaro et al. [6]. The turbines to place are $15MW$. Moreover, the zones have a power density of $4MW/km^2$, which defines the number of turbines to be placed.

Figure 4 shows the available area for instance C from Cazzaro et al. [6].

7.2 Wind

Along with the synthetic sites, wind data [6] is provided from five different sites along the coast of the Netherlands. The data originally stems from the Netherlands Enterprise Agency (RVO) [28] and gives a set of different wind scenarios. Each scenario consists of a wind speed and direction, along with the probability of encompassing the respective scenario. Note that the data from Ten Noorden van de Waddeneilanden has been used for benchmarking in this paper.

7.3 Turbine

Lastly, from Cazzaro et al. [6], is the turbine data, originally from the International Energy Agency, of two different wind turbines, producing $10MW$ [3] and $15MW$ [19] of power. For a number

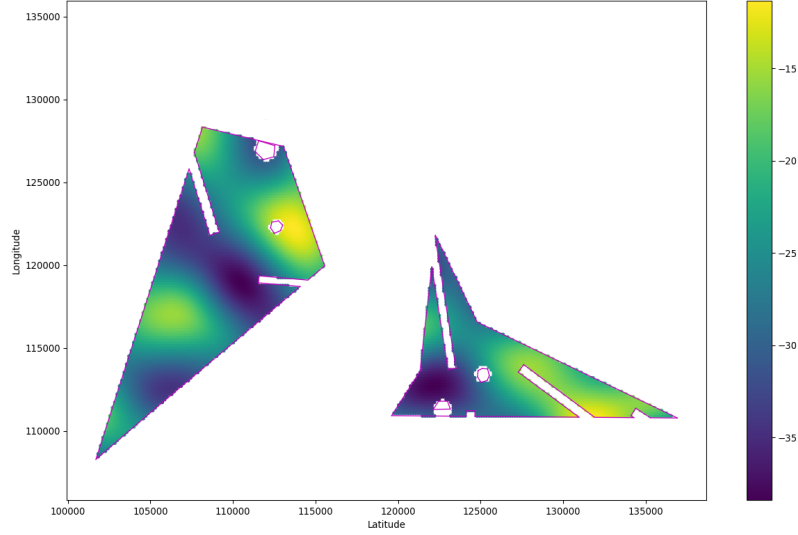


Figure 4: Illustration of Layout C provided by Cazzaro et al. [6]. All non-white areas represent available turbine positions. White regions within the colored areas indicate obstacles. The color bar represents the water dept. Note that neither the substation nor any turbines have been placed.

of different wind speeds (m/s) we get the corresponding power produced (MW) and the thrust coefficient. Note that the turbine producing $15MW$ of power has been used for benchmarking in this project.

The data from the last three sections, the site, wind scenarios and chosen turbine, is needed in order to create the power matrix.

7.4 Power Matrix

In order to generate the power and wake matrix P we used the code by Cazzaro [5]. As input, it takes one of the 10 sites described in subsection 7.1, as well as wind and turbine data, described in subsection 7.2 and subsection 7.3, respectively.

The diagonal of the power matrix, representing the estimated energy production in free wind, is calculated by accounting for the probability distribution of various wind speeds and the corresponding power output of the specific turbine at those speeds. The other elements are found by considering the probability of various wind speeds *and* direction, as well as the distance between turbines. This is then used to compute the wake effect exerted by one turbine on another. The Jensen model [24] was utilized to calculate the wake effect. All values are in megawatts.

Power production is converted to profit using the Energy Price Factor described in the sequel.

7.5 Energy Price Factor (EPF)

In order to determine the full value of a windfarm, one must consider the revenues due to power production within its entire lifetime. The revenue is calculated by multiplying the power production $[MWh/y]$ by the *Energy Price Factor* (EPF) $[EUR/MWh/y]$. The EPF is determined based on

the average energy price in 2021 for Denmark East (DK2), which was 87.72 EUR/MWh [12]. To project total revenue over the 25-year lifespan of the wind farm, an additional adjustment factor is applied when determining the EPF. In this paper, we consider an interest rate of 4.875%, allowing us to calculate the EPF as follows:

$$EPF = 87.72 \sum_{t=1}^{25} \frac{1}{(1 + 0.04875)^t} \text{EUR/MWh/year} \approx 1251 \text{EUR/MWh/year}.$$

With this, we can calculate the revenue, K_{euro} , from element $P_{i,j}$ of the power matrix using the following formula:

$$\text{Revenue}[\text{Euro}] = K_{euro} = P_{i,j} \cdot (365 \cdot 24)[h/y] \cdot EPF[\text{EUR/MWh/y}] = P_{i,j} \cdot 10.9 \text{M€}/\text{MW}.$$

7.6 Cables

This section is concluded with an overview of the cables used for benchmarking. Each cable type corresponds to a specific capacity—indicating the maximum number of turbines it can support—and an associated estimated cost per meter. The cables are summarized in Table 3.

Cable number i	Cable capacity c_i (turbines)	Cable cost a_i (€/m)
1	1	272
2	2	348
3	4	480
4	6	672

Table 3: Overview of cable capacity (max number of supported turbines) and corresponding cost per meter, for the set of cable types C used in the experiments.

8 Computational Experiments

The main goal of this section is to examine the effect of utilizing a semi-integrated approach compared to a sequential approach. For instances A through J, introduced in subsection 7.1, the WFLOP and WFCRP are solved sequentially both with and without an estimate of the cable cost in the WFLOP. This section compares the result both numerically, as well as visually. In addition, we compare our results with state-of-the-art methods. The section concludes with an analysis of the upper and lower bounds presented in subsection 4.1 – examining their performance and quality across the tested instances.

All results were run on an Intel Xeon Gold 6226R Processor [23], using 16 cores. The processor has a cache size of 22MB and a clock frequency of 2.90 GHz. The WFLOP method is implemented in C, while WFCRP is implemented in Julia, using Gurobi version 11.0.2 [21] to solve the MIP model. Each instance in Cazzaro et al. [6] consists of 1 to 3 zones. Our implementation of WFLOP allows solving the full instance at once, i.e., all zones are considered simultaneously. In contrast,

WFCRP processes one zone at a time. WFLOP is run for 2 hours per instance, while WFCRP is allocated a time limit of 2 hours per zone.

8.1 Experimental setup

Recall from subsection 4.2 that an estimate of the cable cost can be added to the diagonal of the modified power matrix Π given by (24). The estimate is defined as a linear combination of the upper and lower bound on cabling costs using weighting $\alpha \in [0, 1]$. A value of $\alpha = 0$ corresponds to the lower bound, while $\alpha = 1$ corresponds to the upper bound. When comparing the performance of the semi-integrated approach with the sequential approach, the semi-integrated approach is run for multiple values of $\alpha = \{0, 0.25, 0.5, 0.75, 1\}$, while the sequential approach, using no cable estimates, use $L^* = \mathbf{0}$.

For each of the instances A through J, the WFLOP is first run without cable estimates, before it is run with cable estimates corresponding to α values of $\{0, 0.25, 0.5, 0.75, 1\}$. Note, whenever the WFLOP is solved with a cable estimate, the cable estimate is originally a negative contribution to the objective value. Hence, we finish by *adding* the cable estimate back to the objective value, such that the final output only equals the revenues from power production, minus construction costs. Moreover, the WFCRP is solved for all instances and values of α .

8.2 Parameter tuning

To optimize model performance, a parameter tuning phase was conducted to identify the most effective hyperparameters for both the initial greedy heuristic and the VNS components of the WFLOP, as well as for the matheuristic algorithm applied to the WFCRP. Details on the parameter tuning can be found in [31]. All chosen parameters are summarized in Table 4.

Init. greedy heur.		VNS for WFLOP			Matheuristic for WFCRP		
n	Pool size	Max-radius	m	Shake-pct	Num-to-move	Group-avg	Group-avg
1	2	4000	128	0.05	0	8	6

Table 4: Summary of parameters for the initial greedy heuristic and VNS solving the WFLOP, as well as the matheuristic solving the WFCRP.

8.3 Effect of parameter α

We begin our analysis by examining how the choice of $\alpha \in [0, 1]$ in (23) influence the results. Intuitively, as the cable estimates increment, the VNS algorithm gives increased consideration to the distance to the substation. As positions far away from the substation are penalized to a greater extent, one would expect turbines in the optimal solution to move closer to the substation, potentially lowering the cable costs. Figure 1 in Section 1 shows the final result both without and with a cable estimate. In figure 1 (left) $\alpha = 0$, hence no cable estimates are taken into account,

while in figure 1 (right) $\alpha = 1$ is used. Comparing the two solutions it can be seen that turbines generally move closer to the substation once a cable estimate is added.

The decreasing revenues from energy production and the corresponding reduction in cable cost with increasing α is seen in Figure 5. Plotting the two bar graphs on the same scale shows that the relative performance of the WFLOP does not decrease nearly as much as the cable costs. That said, since cable costs are typically 10–100 times lower than the estimated profits from power production, as can be seen in the full results in Appendix B.1, even a small relative decrease in power output can result in a large absolute drop in the total objective value—often outweighing any gains from reduced cable costs.

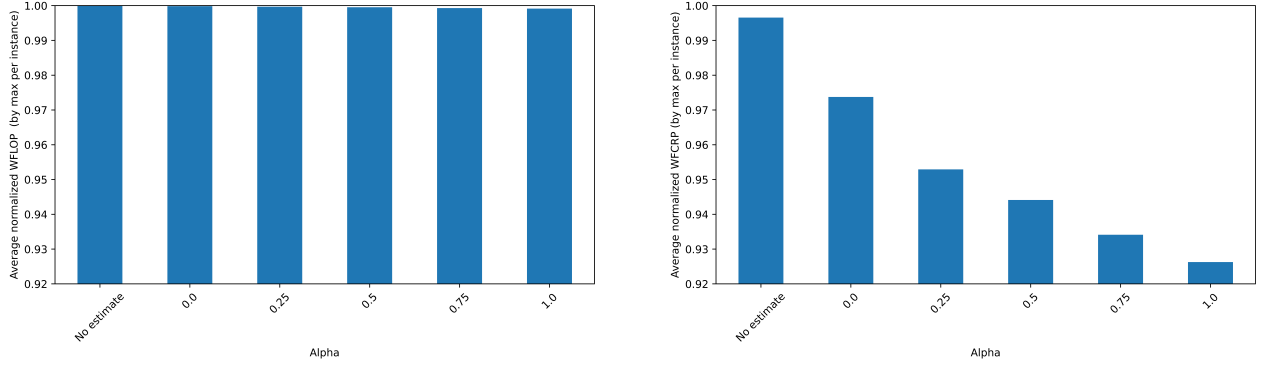


Figure 5: (left) The average ratio between the WFLOP objective and the best-known objective for each value of α . (right) The average ratio between the WFCRP objective and the maximum cable cost for each value of α . The bar graph values are reported in Appendix B.2.

Next, Figure 6 (left) illustrates the relative difference between the estimated cable costs in the WFLOP and actual cable costs, while Figure 6 (right) presents, for a given α , the following: for each instance we find the best-known integrated objective value (z_{int}^{best}) and calculate the ratio between z_{int}^α (the integrated objective value given α) and z_{int}^{best} . Finally, an average across all instances is taken. From Figure 6 (left), it is evident that $\alpha = 0.25$ yields, on average, the most accurate cable cost estimates. Consistently, Figure 6 (right) indicates that this value of α also results, on average, in the best overall performance. These findings suggest that incorporating more accurate cable cost estimates into the WFLOP may improve the overall solution quality.

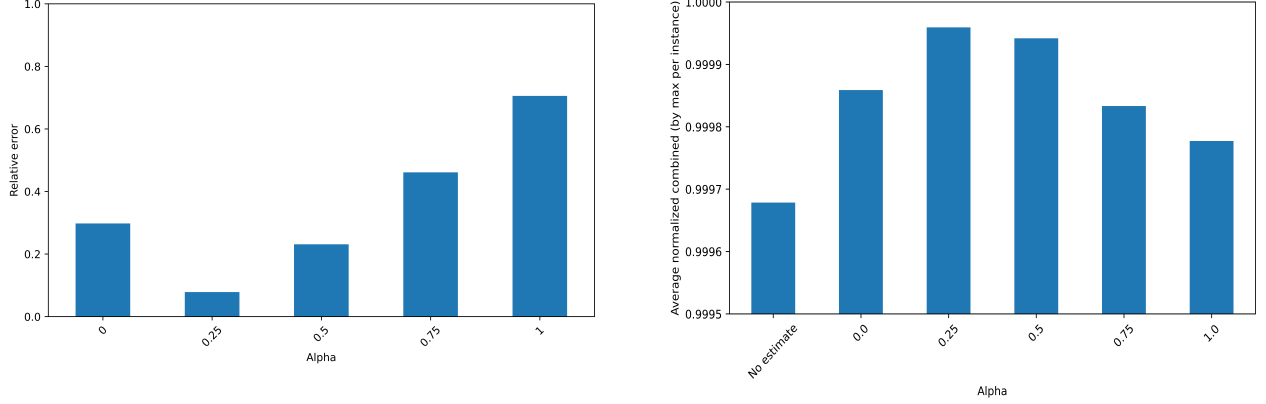


Figure 6: (left) The average relative error in cable costs for each value of α . (right) The average ratio between combined objective value and the best-known objective for each value of α . The actual values are reported in Appendix B.2.

8.4 Sequential vs. semi-integrated approach

We proceed by comparing the performance of the purely sequential method, i.e. no cable estimate in the WFLOP, with the semi-integrated method. Note, the result of the semi-integrated approach corresponds to the maximum across all values of α . Table 5 shows the relative decrease in the WFLOP and WFCRP when moving from a sequential to a semi-integrated approach. It can be seen from the two middle columns that the relative decrease in the WFLOP is much lower than the relative decrease in the cable costs. The total savings in each instance is reported in the last column. We see an average saving of more than 2 million euros. A detailed summary of all results can be found in Appendix B.1.

It is important to notice that cable costs should be paid immediately as the wind farm is built, while revenues from power production are estimates that may take years to materialize. Hence, saving millions of euros during the construction phase can be highly beneficial for companies, as it lowers the barrier to entry, and the importance should therefore not be underrated.

8.5 Comparison to existing Approaches

We next compare our results with existing solution methods, particularly the combined approach presented by Cazzaro et al. [10], as the authors’ method was tested on the same instances as we study. This is done by adjusting our EPF and cable costs to those used by Cazzaro et al. [10] and rerunning our experiments. Table 6 demonstrates that our approach achieves higher objective values compared to those reported in [10]. However, due to differences in the problem setup, a fully reliable comparison is not possible. Notably, Cazzaro allows for multiple base stations in some instances, whereas our formulation uses only one per zone. Additionally, certain parameters—such as the power matrix—may differ slightly between the two models due to different implementations of wake calculation.

Instance	Percentage decrease WFLOP	Percentage decrease WFCRP	Savings (Euro)
A	0.035 %	9.2 %	1 140 000
B	0.003 %	0.6 %	235 000
C	0.029 %	4.1 %	700 000
D	0.030 %	4.2 %	3 610 000
E	0.028 %	12.4 %	9 225 000
F	0.024 %	2.3 %	550 000
G	0.026 %	3.8 %	1 580 000
H	0.012 %	1.0 %	80 000
I	0.027 %	1.7 %	15 000
J	0.027 %	4.4 %	3 010 000
Average	0.024 %	4.3 %	2 015 000
Median	0.027%	3.95%	920,000

Table 5: Overall comparison of the performance of the purely sequential method with the semi-integrated method for instances A through J from Table 2.

Instance	Cazzaro et al. (M€)	Semi-integrated method (M€)
A	685	701
B	1625	1658
C	1349	1373
D	2745	2795
E	2270	2312
F	2609	2657
G	2290	2341
H	3070	3146
I	4956	5041
J	3799	3920

Table 6: Comparison of objective values between Cazzaro et al. [10] and the semi-integrated method (in million euros).

8.6 Bounds

We now turn our attention to the quality of the bounds presented in Section 4, beginning with their performance on the integrated problem. Table 7 reports the relative gap between the upper and lower bounds, both for the polynomial bound, as well as after applying the solution method described in Section 5. On average, across the ten instances, the relative gap equals 7.31% and 5.57% for the polynomial bound and the bound after running the model, respectively. Computing the gap (i.e. both upper and lower bounds) takes approximately 1 second for the smallest instance and 5 minutes for the largest instance.

We conclude the results by assessing the significance of using Corollary 1 to lift the initial lower bound, $\mathcal{L}_1$, on the cabling cost. Table 8 reports the initial lower bound and the lower bound after applying Corollary 1, i.e. $\mathcal{L}_5$, as well as the relative increase in the lower bound. The upper bound is the cable cost after running the MIP model. On average, we observe a relative increase of 5.93%

Instance	Relative gap polynomial bounds	Relative gap after running model
A	3.01 %	2.27 %
B	6.19 %	4.98 %
C	5.33 %	3.86 %
D	9.67 %	7.56 %
E	6.89 %	4.55 %
F	7.26 %	5.37 %
G	7.23 %	5.93 %
H	7.89 %	6.20 %
I	11.14 %	9.09 %
J	8.49 %	5.95 %
Average gap	7.31 %	5.57 %

Table 7: The relative gap between $\mathcal{U}_4^{int}$ and $\mathcal{L}^{int}$. In both cases (column 2 and 3) the $\mathcal{U}_4^{int}$ is found using the $O(n^2)$ upper bound, $\mathcal{U}_1$ [29]. For the NP bound (column 2), $\mathcal{L}^{int}$ is found using the $O(n^3)$ greedy heuristic described in Section 5.1.1. For the last column, $\mathcal{L}^{int}$ is the result of running the solution method described in Section 5 on the integrated model.

when applying Corollary 1. In some cases the increase is over 11% showing the strength of the lifting approach. The detailed results of $\mathcal{L}_1$, $\mathcal{L}_2$, $\mathcal{L}_3$ and $\mathcal{L}_4$ are shown in Appendix B.3. In the results presented in this paper, we observe that $\mathcal{L}_5 = \mathcal{L}_3$ in all instances. However, this does not necessarily hold in general, as demonstrated by the additional results provided in Appendix B.3.

9 Conclusion

Given the ongoing challenges resulting from climate change and rising global demand for sustainable energy, decreasing the cost of renewable energy is vital for the green transition. Traditionally, wind farms were designed by first solving the WFLOP and then WFCRP, but an integrated approach may lead to more economically efficient wind farm designs.

In this paper we presented the first efficient (polynomial) upper and lower bounds for the integrated problem, and showed that they are relatively tight. The integrated bounds make use of anticipated cabling costs to each of the possible positions, controlled by a parameter $\alpha \in [0, 1]$. Using $\alpha = 1$ we get upper bounds on the integrated problem, while using $\alpha = 0$ we get lower bounds on the same problem.

Finally, we used the anticipated cabling costs in a semi-integrated model, where we first solved the WFLOP using a proper value of α , and then solved the WFCRP for the selected positions. The computational results show that the proposed approach is able to find better solutions than the state-of-the-art algorithm by Cazzaro [10]. Furthermore, we showed that the semi-integrated approach saves around 4% in cable costs compared to a stepwise approach, leading to average savings of more than 2 million euro per instance.

In contrast to the branch-and-cut algorithm proposed by Fischetti et al. [16], which was limited to instances with up to 100 available positions, our method demonstrates significantly greater

Instance / zone	$\mathcal{U}$ (M€)	$\mathcal{L}_1$ (M€)	$\mathcal{L}_5$ (M€)	Relative increase
A / 0	12.76	7.42	8.29	6.84 %
A / 1	9.89	3.37	3.84	4.76 %
B / 0	64.66	45.93	49.22	5.09 %
C / 0	34.05	22.69	24.71	5.92 %
C / 1	11.79	5.74	6.76	8.55 %
D / 0	145.58	118.01	123.72	3.92 %
E / 0	1.82	0.79	0.99	11.07 %
E / 1	68.57	52.30	55.46	4.61 %
E / 2	18.67	11.20	12.41	6.48 %
F / 0	90.14	67.12	71.55	4.92 %
F / 1	14.21	8.75	9.63	6.15 %
G / 0	88.85	63.82	68.52	5.29 %
H / 0	106.61	81.51	86.82	4.98 %
H / 1	16.60	9.93	10.94	6.07 %
I / 0	256.32	200.81	211.29	4.09 %
J / 0	87.68	65.03	69.60	5.21 %
J / 1	44.32	31.16	33.65	5.61 %
J / 2	11.37	5.67	6.50	7.09 %
Average				5.93 %

Table 8: For each zone the upper bound (solution from running MIP model) and initial lower bound is reported, alongside the lifted lower bound, $\mathcal{L}_5$. Finally, we calculate the relative increase in the lower bound using $\mathcal{L}_5$. The relative increase is taken with respect to the upper bound.

scalability, having been applied successfully to instances with over 21,000 available positions, similar to the work by Cazzaro et al. [10].

The core advantage of the proposed method lies in its simplicity and compatibility. Since only one of the inputs to the WFLOP is modified, the semi-integrated approach enables the use of existing, state-of-the-art methods for both the WFLOP and WFCRP. This means that, if a company already has implementations of the two solution methods, shifting from a sequential to a semi-integrated approach requires minimal changes. Yet, the potential savings are substantial, reducing upfront investments, and hence also risks. Furthermore, future improvements in solving the two underlying problems WFLOP and WFCRP can immediately be transferred to the semi-integrated approach.

9.1 Future work

As noted in Corollary 7 a variety of upper bounds for PDSP can be used to bound the integrated WFLOP and WFCRP. In our experiments we only used a simple upper bound for PDSP, hence it would be interesting to study how much the overall bound for IWFLOP can be improved by using tighter bounds for PDSP. It would also be interesting to use the polynomial bounds for the integrated problem in an exact algorithm.

Future work includes refining the cable cost estimates by accounting for the relative positions of turbines — basing the estimate on turbine-to-turbine relationships rather than treating each turbine independently.

Finally, several approximation algorithms exist for the PDSP, hence it would be interesting to study whether they can be generalized to the integrated WFLOP and WFCRP problem. This would lead to polynomial algorithms with a performance guarantee. Although such a result will be mainly theoretically interesting, it may inspire development of practical solution methods also.

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CRedit author statement

The authors declare that they have contributed equally.

Code and Data

Will be made available on SSRN after blind review. A link will be inserted in this section.

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A Appendix: Proofs of bounds

A.1 Proof: Proposition 1

We begin by defining a lemma needed for the proof.

Lemma A: *For any turbine $i \in T$, let p_i denote the shortest path to the substation, with length d_i . Then p_i and p_j need not intersect for any turbines $i, j \in T$.*

Proof. Assume for the sake of contradiction that there exists two turbines $i, j \in T$ such that p_i and p_j intersect at some point r . Now, divide p_i and p_j into two sections, the one before they reach r and the one after. That is, for turbine i , we define p_i^{r-} (before r) and p_i^{r+} (after r), with distances d_i^{r-} and d_i^{r+} . Similar variables are defined for turbine j .

First, consider the case where $d_i^{r+} = d_j^{r+}$. Then we can simply set $p_i^{r+} = p_j^{r+}$, keeping the distance constant and avoiding cable crossings by allowing the cables to run in parallel. Second, consider the case where $d_i^{r+} < d_j^{r+}$. Then the path between $j \in T$ and the substation, defined by p_j^{r-} followed by p_i^{r+} would be shorter than p_j . This is a contradiction, as p_j is the shortest path from j to the substation. Similarly, $d_i^{r+} > d_j^{r+}$ is a contradiction. Hence, the shortest paths p_i and p_j need not intersect. $\square$

We proceed with the proof of Proposition 1.

Proof. We prove this by constructing a feasible solution with an objective value of $\sum_{i \in T} d_i a_1$. This implies that the objective value is an upper bound to the cabling problem.

For all turbines $i \in T$, place a cable of cost a_1 per unit along the shortest path of length d_i . The total cable cost will then equal $\sum_{i \in T} d_i a_1$, as desired. By Lemma A, there will be no cable intersections. Moreover, all other constraints defined in subsection 3.2 are satisfied, meaning that we have a feasible solution. $\square$

A.2 Proof: Proposition 2

We begin by defining a lemma needed for the proof.

Lemma B: *Assume f is a cable leaving the substation, and T_f is the set of turbines connected to cable f . Then*

$$\sum_{t \in T_f} d_t \frac{a_n}{c_n}$$

is a lower bound on the cost of cable f .

Proof. Assume we have a set of edges, E , connecting the turbines in T_f to the substation. We continue by dividing the edges into groups based on the number of turbines they support, $e_1^1, \dots, e_{r_1}^1, e_1^2, \dots, e_{r_2}^2, \dots, e_1^n, \dots, e_{r_n}^n$, where e_j^i is the j -th cable (in no particular order) of edges supporting at most c_i turbines. Here, r_i equals the number of edges in group i , i.e. the edges that support at most c_i turbines.

With $d_{e_j^i}$ equal to the distance of edge e_j^i , the total cost of the cable in an optimal solution equals

$$\text{cost}_f = a_1(d_{e_1^1} + \dots + d_{e_{r_1}^1}) + a_2(d_{e_1^2} + \dots + d_{e_{r_2}^2}) + \dots + a_n(d_{e_1^n} + \dots + d_{e_{r_n}^n}).$$

Moreover, due to economy of scale, we have

$$\begin{aligned} \text{cost}_f &= a_1(d_{e_1^1} + \dots + d_{e_{r_1}^1}) + a_2(d_{e_1^2} + \dots + d_{e_{r_2}^2}) + \dots + a_n(d_{e_1^n} + \dots + d_{e_{r_n}^n}) \\ &\geq \frac{c_1 a_n}{c_n}(d_{e_1^1} + \dots + d_{e_{r_1}^1}) + \frac{c_2 a_n}{c_n}(d_{e_1^2} + \dots + d_{e_{r_2}^2}) + \dots + \frac{c_n a_n}{c_n}(d_{e_1^n} + \dots + d_{e_{r_n}^n}) \\ &= \frac{a_n}{c_n}(c_1 d_{e_1^1} + \dots + c_1 d_{e_{r_1}^1}) + \frac{a_n}{c_n}(c_2 d_{e_1^2} + \dots + c_2 d_{e_{r_2}^2}) + \dots + \frac{a_n}{c_n}(c_n d_{e_1^n} + \dots + c_n d_{e_{r_n}^n}) \\ &= \frac{a_n}{c_n} \left(c_1 d_{e_1^1} + \dots + c_1 d_{e_{r_1}^1} + c_2 d_{e_1^2} + \dots + c_2 d_{e_{r_2}^2} + \dots + c_n d_{e_1^n} + \dots + c_n d_{e_{r_n}^n} \right). \end{aligned}$$

Further, the cable path between any turbine $t_i \in T_f$ and the substation is a combination of edges. Moreover, at most c_i turbines can use an edge of type e_j^i , for any $j \in [1, \dots, r_i]$. Finally, the path from t_i to the substation along the cable is at least the length of the shortest distance to the substation, d_{t_i} . Mathematically, for any turbine $t \in T_f$, we have $\sum_{e \in \text{Path}_t} d_e \geq d_t$, where Path_{t_i} is the set of edges in the path for turbine t to the substation, following the edges of cable f . Hence,

$$\begin{aligned} \text{cost}_f &\geq \frac{a_n}{c_n} \left(c_1 d_{e_1^1} + \dots + c_1 d_{e_{r_1}^1} + c_2 d_{e_1^2} + \dots + c_2 d_{e_{r_2}^2} + \dots + c_n d_{e_1^n} + \dots + c_n d_{e_{r_n}^n} \right) \\ &\geq \frac{a_n}{c_n} \sum_{t \in T_f} d_t, \end{aligned}$$

as desired. Note that the second inequality comes from the fact that for any edge e_j^i , the number of turbines $t \in T_f$ such that $e_j^i \in \text{Path}_t$ does not exceed c_i . $\square$

We proceed with the proof of Proposition 2.

Proof. Assume we have an arbitrary solution to the WFCRP and define F as the set of cables leaving the substation. By using Lemma B and summing over all cables leaving the substation, we get a total cable cost of

$$\sum_{f \in F} \text{cost}_f \geq \sum_{f \in F} \frac{a_n}{c_n} \left(\sum_{t \in T_f} d_t \right) = \sum_{i \in T} d_i \frac{a_n}{c_n},$$

as desired. Note that the equality stems from the fact that each turbine is associated with one, and only one, cable $f \in F$. $\square$

A.3 Proof: Proposition 3

We begin by showing how the lower bound can be lifted in the case where the cabling should satisfy a string topology, i.e. each turbine may have at most 1 ingoing cable, before generalizing to a branched topology, where turbines may have more than 1 ingoing cable.

Lemma C: *Assume that the cabling should satisfy a string topology. Then we know that at least $H = \lceil \frac{q}{c_n} \rceil$ strings will be used. Moreover, assume that the distance to the substation exceeds δ for all turbines. Then*

$$H\delta \left(a_1 - \frac{a_n}{c_n} \right) + \sum_{i \in T} d_i \frac{a_n}{c_n}$$

is a lower bound on the total cable cost.

Proof. At the end of each string there is only 1 turbine. The cable between this turbine and the next turbine (or substation) must have a minimum length of δ . Moreover, the cheapest cable, with cost a_1 , would be chosen in the optimal solution. Thus, with at least H strings, we can add $H\delta a_1$ to $\mathcal{L}_1$. Now, for the turbines at the end of the string, we must consider the fact that when creating $\mathcal{L}_1$ we added a contribution of $d_i \frac{a_n}{c_n}$ from the turbine, where d_i is the shortest path from the turbine to the substation. It may be the case that the cable of length δ and cost a_1 overlaps with the shortest path. Hence, we must compensate for this by subtracting $\delta \frac{a_n}{c_n}$ from the contribution of the turbine to ensure it remains a valid lower bound. In short, when adding $H\delta a_1$, we must also subtract $H\delta \frac{a_n}{c_n}$. $\square$

Further, Lemma C can be extended to the case where all turbines are located a minimum distance of $g\delta$ from the substation, where g is an integer such that $g \leq c_1$.

Lemma D: *Assume that the cabling should satisfy a string topology. Then we know that at least $H = \lceil \frac{q}{c_n} \rceil$ strings will be used. Moreover, assume that the distance to the substation exceeds $g\delta$ for all turbines, where $g \in \mathbb{N}$ and $g \leq c_1$. Then*

$$H\delta \left(ga_1 - \frac{a_n}{c_n} \sum_{i=1}^g i \right) + \sum_{i \in T} d_i \frac{a_n}{c_n}$$

is a lower bound on the total cable cost.

The idea behind the lemma is based on the observation that the end segment (farthest from the substation) of the string must have a length of at least $g\delta$, as the distance to the substation exceeds $g\delta$ for all turbines. Moreover, as the minimum distance between turbines is at least δ , at most g turbines can be supported by the end segment of length $g\delta$. An optimal solution would use the cheapest cable of cost a_1 , as $g < c_1$. This allows us to add an additional term, $g\delta a_1$ to account for the cost of this initial stretch. That being said, when creating the lower bound in Proposition 2 we added a value of $\delta \frac{a_n}{c_n}$ to compensate for the first turbine. However, it is possible that the turbines are arranged in a straight line from the substation. Then the initial segment of length δ from the second turbine is compensated by $\delta \frac{a_n}{c_n}$ from both the second *and* the first turbine, i.e. $2\delta \frac{a_n}{c_n}$ in the

lower bound. An illustration is shown in Figure 7. This must therefore be subtracted from the lower bound when adding $g\delta a_1$. We proceed with the formal proof.

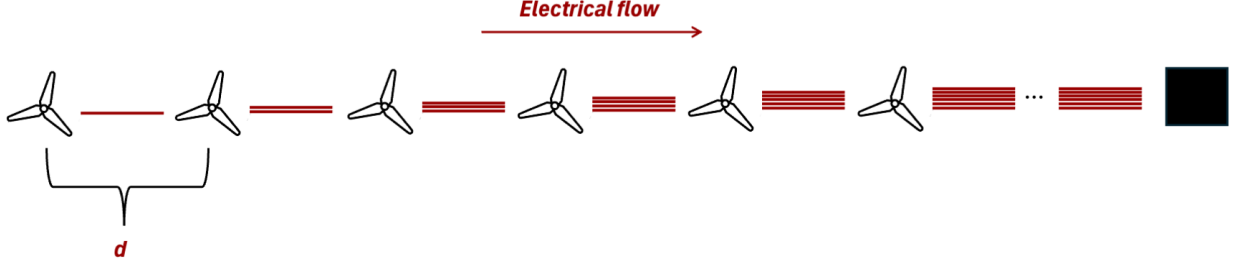


Figure 7: Example illustration supporting proof of Lemma D. The red lines illustrate the total contribution to $\mathcal{L}_1$ due to the six turbines. Given that the most cost efficient cable supports 6 turbines, the distance between the left-most and second-left-most turbine adds $\delta \frac{a_n}{6}$ to $\mathcal{L}_1$, while the next segment (second-left-most to third-left-most turbine) adds $2\delta \frac{a_n}{6}$ to the lower bound, and so on. The black square represents the substation.

Proof. For at least H strings, we have one out of two cases.

Case 1: At the end of the string there will be at least g turbines connected in sequence. The cable connecting these g turbines to each other and the next turbine (or substation) must have a length of at least $g\delta$.

Case 2: There will be less than g turbines connected. Since all turbines are at least $g\delta$ away from the substation, this cable must have a length of at least $g\delta$.

In both cases we have a cable with g or less turbines connected to it, with a length equal to at least $g\delta$. In an optimal solution, the cost of this cable is a_1 , i.e. the cheapest cable cost.

Given an arbitrary string, we set r equal to the total number of turbines on the string. Moreover, set $f = \min(r, g)$. Assume $t_1, t_2, \dots, t_f, t_{f+1} + \dots + t_r$ are the turbines on the string, where t_1 is the last turbine (furthest from the substation), t_2 is the second to last, etc.

We proceed by placing the cable of length $g\delta$ and cost a_1 before showing that the cost of the remaining string cannot exceed

$$\frac{a_n}{c_n} \left(\sum_{i=1}^r d_{t_i} - \delta \sum_{i=1}^g i \right).$$

Starting at turbine t_1 we place the cheapest cable until we reach the next turbine, or the total distance of the cable equals $g\delta$. With the remaining cable, we do the same from t_2, t_3 , etc. As turbines must be a distance δ apart, the last turbine considered cannot exceed t_f . Let p^* be the position where we have reached a cable length of $g\delta$. Assume p^* is located on the edge leaving t_l , where $1 \leq l \leq f$. Then the remaining cable, must cover $p^*, t_{l+1}, \dots, t_r$ before reaching the substation.

Furthermore, we associate all turbines t_i such that $i \leq l$, i.e. the turbines considered before reaching a total cable length of $g\delta$, with position p^* . Then, the remaining cable must cover $p^*, \dots, p^*, t_{l+1}, \dots, t_r$, which is identical to $p^*, t_{l+1}, \dots, t_r$, where there are multiple edges of length

0 and capacity c_1 connecting p^* to itself. This allows us to use Lemma B. By Lemma B,

$$\frac{a_n}{c_n} \left(l \cdot d_{p^*} + \sum_{i=l+1}^r d_i \right)$$

is a lower bound on the remaining cable cost. Furthermore, p^* must be a minimum of

$$\max(d_{t_1} - g\delta, d_{t_2} - (g-1)\delta, \dots, d_{t_l} - (g-l+1)\delta)$$

units from the substation. Hence,

$$\begin{aligned} & \frac{a_n}{c_n} \left(l \cdot d_{p^*} + \sum_{i=l+1}^r d_i \right) \\ & \geq \frac{a_n}{c_n} \left((d_{t_1} - g\delta) + (d_{t_2} - (g-1)\delta) + \dots + (d_{t_l} - (g-l+1)\delta) + \sum_{i=l+1}^r d_i \right) \\ & = \frac{a_n}{c_n} \left(\sum_{i=1}^r d_{t_i} - \delta \sum_{i=g-l+1}^g i \right) \\ & \geq \frac{a_n}{c_n} \left(\sum_{i=1}^r d_{t_i} - \delta \sum_{i=1}^g i \right). \end{aligned}$$

Therefore, the total cost of the string is no greater than

$$\begin{aligned} & g\delta a_1 + \frac{a_n}{c_n} \left(\sum_{i=1}^r d_{t_i} - \delta \sum_{i=1}^g i \right) \\ & = \frac{a_n}{c_n} \sum_{i=1}^r d_{t_i} + \delta \left(ga_1 - \frac{a_n}{c_n} \sum_{i=1}^g i \right). \end{aligned}$$

Note, the term $\frac{a_n}{c_n} \sum_{i=1}^r d_{t_i}$ is the contribution from the turbines in $\mathcal{L}_1$. In other words, for each string, we can add $\delta(ga_1 - \frac{a_n}{c_n} \sum_{i=1}^g i)$ to $\mathcal{L}_1$. As there are at least H strings, we can add $H\delta(ga_1 - \frac{a_n}{c_n} \sum_{i=1}^g i)$. $\square$

We proceed by extending the lemma to cases where some turbines are located close to the substation.

Lemma E: *Assume that the cabling should satisfy a string topology. Moreover, assume that the distance to the substation exceeds $g\delta$ for all but m turbines, where $g \in \mathbb{N}$ and $g \leq c_1$. Define $G = \lceil \frac{g-m}{c_n} \rceil$. Then*

$$G\delta \left(ga_1 - \frac{a_n}{c_n} \sum_{i=1}^g i \right) + \sum_{i \in T} d_i \frac{a_n}{c_n}$$

is a lower bound on the total cable cost.

Proof. There are $q - m$ turbines located a distance of at least $g\delta$ from the substation. Hence, there must be at least $G = \lceil \frac{q-m}{c_n} \rceil$ strings that connect them to the substation.

For each of the G strings, we have one of two cases.

Case 1: At the end of the string there will be g turbines connected in sequence. The cable connecting these g turbines to each other and the next turbine (or substation) must have a length of at least $g * \delta$. This is because the g turbines must be a distance δ apart. Moreover, at least one of the turbines is more than $g\delta$ from the substation, meaning that if the last of the g turbines is connected to the substation with a cable shorter than δ , then this distance has been compensated for by the fact that at least one turbine is greater than $g\delta$ away.

Case 2: There will be less than g turbines connected. Since at least one turbine is a distance of $g\delta$ from the substation, this cable must have a length of at least $g\delta$.

The remainder of the proof is the same as the proof of Lemma D, after the two cases are defined. Thus, we multiply $\delta \left(ga_1 - \frac{a_n}{c_n} \sum_{i=1}^g i \right)$ by G , and add it to $\mathcal{L}_1$. $\square$

The final step is to extend Lemma E to a branched topology, where a turbine can have more than 1 ingoing cable. This leads us to Proposition 3.

Proposition 3: *Assume that the cabling should satisfy a branched topology. Moreover, assume that the distance to the substation exceeds $g\delta$ for all but m turbines, where $g \in \mathbb{N}$ and $g \leq c_1$. Define $G = \lceil \frac{q-m}{c_n} \rceil$. Then*

$$\mathcal{L}_2 = Gd \left(ga_1 - \frac{a_n}{c_n} \sum_{i=1}^g i \right) + \sum_{i \in T} d_i \frac{a_n}{c_n}$$

is a lower bound on the total cable cost.

Proof. Although we allow for a branched topology, at least $G = \lceil \frac{q-m}{c_n} \rceil$ wires will be connected to the substation with at least one turbine located a distance greater than $g\delta$ from the substation. For each of the G strings, we have one of four cases.

Case 1 (similar to stringed topology): At the end of the cable there are at least g turbines connected in sequence. The cable connecting these g turbines to each other and the next turbine (or substation) must have a length of at least $g\delta$, since at least one turbine is more than $g\delta$ from the substation.

Case 2: There are less than g turbines connected. Since at least one turbine is more than $g\delta$ units away from the substation, the total cable must have a length of at least $g\delta$.

Case 3: There are g turbines connected and they do not form a stringed topology. In this case, each of the g turbines must be connected with a cable of length δ to another turbine or the substation. Since at least one turbine is more than $g\delta$ units away from the substation, the total length must equal at least $g\delta$.

Case 4: There are more than g turbines connected and we have a branching topology. We will now show how to locate g turbines that are connected by the cheapest wire. Starting at the substation, we do a depth-first search along the wire. Consider a tree starting at the first turbine connected to the substation and extending outward. At each turbine we choose to move to a turbine

further down in the tree. We choose the next turbine by following one of the non-cheapest cables. That is, we only move between turbine $t_1 \in T$ and $t_2 \in T$ if they are connected by a cable $i \in C$ where $i \neq 1$. We continue until the current node is only connected to turbines using cable type $1 \in C$. There must be at least g turbines connected to this node and they must all be connected with a cable of length δ to another turbine. This gives a total length of at least $g\delta$. Note that we must have at least g turbines connected to the current node, because if there were less, the optimal solution would have a cable of type $1 \in C$ going from the current node to the previous node. This would be a contradiction.

For case 1, the remainder of the proof is the same as the proof of Lemma D after the two cases are defined, with G instead of H strings.

For cases 2, 3 and 4 we consider a tree starting at the first turbine from the substation and extending outward. Assuming there are r turbines, we want to order them $t_1, t_2, \dots, t_r$. Define the depth of a turbine as the number of turbines separating it from the substation. The deepest turbine is the turbine with the highest number of turbines separating it from the substation. The turbines are then ordered such that the deepest turbines are given the lowest order, i.e. t_1 . The order of turbines at the same depth is arbitrary. The remainder of the proof is the same as the proof of Lemma D, after the turbines are ordered.

□

A.4 Proof: Lemma 1

Proof. Choose an arbitrary i such that $1 \leq i \leq c_n$. Then, the entering cable supports at least $i - 1$ turbines and the exiting cable supports at least i turbines. Hence, within a radius $\frac{1}{2}\delta$ of the turbine we must pay $\frac{1}{2}\delta a_h$ for the ingoing and $\frac{1}{2}\delta a_j$ for the outgoing cable (given the definition of a_h and a_j described in Lemma 1). Meanwhile, in an ideal scenario with no cable loss, as shown in Proposition 2, we would pay $\frac{1}{2}\delta(i - 1)\frac{a_n}{c_n}$ for the ingoing and $\frac{1}{2}\delta i\frac{a_n}{c_n}$ for the outgoing cable. Hence, $\ell_i^s = \frac{1}{2}\delta \left(a_h + a_j - ((i - 1) + i)\frac{a_n}{c_n} \right)$, i.e. the difference between the two, is in fact the loss due to imperfect cabling.

□

A.5 Proof: Lemma 2

Proof. If there are zero turbines to support, then no cable is needed, i.e. $g_0 = 0$. Now, let $j \geq 1$ be arbitrary and assume that for all $j' < j$, we know that $g_{j'}$, as defined, is in fact the minimum cost of supporting j' turbines. We want to show that g_j , as defined, is the minimum cable cost supporting j turbines. Let $i \in C$ be arbitrary. Then we know there exists a set of cables with cost $a_i + g_{\max(0, j - c_i)}$ supporting j turbines, as a_i covers the cost of c_i turbines and there exists a grouping of turbines with cost $g_{\max(0, j - c_i)}$ supporting $\max(0, j - c_i)$ turbines, by our initial assumption. As $i \in C$ was arbitrary, we know that $g_j = \min_{i \in C} (a_i + g_{\max(0, j - c_i)})$ is an upper bound on the minimum cable cost. Assume for the sake of contradiction that there exists a grouping, C_j (some cables may be repeated), of cables from C supporting at least j turbines and has a lower cost, $g^* < g_j$. Let $l \in C_j$

be arbitrary. Then, for some $b \in \mathbb{R}$,

$$g_* = a_l + b < \min_{i \in C} (a_i + g_{\max(0, j - c_l)}) \leq a_l + g_{\max(0, j - c_l)}.$$

In other words, $b < g_{\max(0, j - c_l)}$ and it must support at least $\max(0, j - c_l)$ turbines, meaning we have found a cheaper way of supporting $\max(0, j - c_l)$ turbines. This is a contradiction of our preliminary assumption. Hence, we may conclude that g_j as defined in fact is the minimum cable cost supporting j turbines.

By strong induction, it holds for all j such that $0 \leq j$. $\square$

A.6 Proof: Lemma 3

Proof. Choose an arbitrary i such that $1 \leq i \leq c_n$. Then, the entering cables supports at least $i - 1$ turbines and has a minimum cost of g_{i-1} by Lemma 2 and the exiting cable supports at least i turbines with minimum cost a_h . Hence, within a radius $\frac{1}{2}\delta$ of the turbine we must pay $\frac{1}{2}\delta g_{i-1}$ for the ingoing cables and $\frac{1}{2}\delta a_h$ for the outgoing cable. Meanwhile, in an ideal scenario with no cable loss, as shown in Proposition 2, we would pay $\frac{1}{2}\delta(i - 1)\frac{a_n}{c_n}$ for the ingoing and $\frac{1}{2}\delta i\frac{a_n}{c_n}$ for the outgoing cable. Hence, $\ell_i^b = \frac{1}{2}\delta \left(g_{i-1} + a_h - ((i - 1) + i)\frac{a_n}{c_n} \right)$, i.e. the difference between the two, is in fact the loss due to imperfect cabling. $\square$

A.7 Proof: Proposition 4

Proof. Proposition 2 gives the most cost-efficient lower bound, where there is no waste due to suboptimal cabling. On the other hand, $l_{\min}$ is the minimum amount that is paid due to imperfect cabling at every turbine located at least $\frac{1}{2}\delta$ from the substation. As there are $q - m$ turbines where we waste at least $l_{\min}$, we can add $(q - m)l_{\min}$ to $\mathcal{L}_2$. $\square$

A.8 Proof: Proposition 5

Proof. All q wind turbines must have a cable connecting them to another turbine or to the substation. As turbines are at least a distance δ apart, the cables between turbines must have a length of at least δ . Moreover, all cables going from a turbine to the substation will have a length of at least δ for all turbines placed more than a distance δ away from the substation. As there are at most m turbines placed within a distance δ of the substation, there will be at least $q - m$ cables of at least length δ (either between turbines or to the substation).

Finally, the minimum cost of a cable is a_1 per distance unit. Hence, the cable cost will be at least $(q - m)\delta a_1$. $\square$

A.9 Proof: Proposition 6

Proof. Let s_{int} be an arbitrary solution to the QWFP and let x be the associated solution vector. Let x be the solution vector to the WFLOP and connect all turbines to the cheapest substation

using the cheapest cable with cost a_1 . The objective value of the integrated problem is then as follows:

$$\begin{aligned}
z_{int} &= z_{layout} - z_{cable} \\
&= \sum_{i \in N} \sum_{j \in N} R_{i,j} x_i x_j - \sum_{i \in N} x_i d_i a_i \\
&= \sum_{i \in N} \sum_{j \in N} R_{i,j} x_i x_j - \sum_{i \in N} x_i (l_i + 1 \cdot (u_i - l_i)) \\
&= \sum_{i \in N} x_i \left(\sum_{j \in N} (R_{i,j} x_j) - L_i^1 \right),
\end{aligned}$$

final equation is equivalent to redefining Π using (24) with $\alpha = 1$ and solving the QWFP.

Hence, for any solution to the QWFP with Π redefined using (24) with $\alpha = 1$ we can find a corresponding solution to the IWFLOP with the same objective value. Hence, solving the QWFP with Π redefined using (24) with $\alpha = 1$ is a lower bound on the IWFLOP. $\square$

A.10 Proof: Proposition 7

Proof. Let s_{int} be an arbitrary solution to the integrated problem, with objective value z_{int} . Let z_{layout} and z_{cable} be the associated objective values for the WFLOP and WFCRP, respectively. Moreover, assume x is the solution vector of the WFLOP associated with solution s_{int} . Then,

$$\begin{aligned}
z_{int} &= z_{layout} - z_{cable} \\
&\leq z_{layout} - \sum_{i \in N} x_i d_i \frac{a_n}{c_n} \\
&= \sum_{i \in N} \sum_{j \in N} R_{i,j} x_i x_j - \sum_{i \in N} x_i (\ell_i + 0 \cdot (u_i - \ell_i)) \\
&= \sum_{i \in N} \sum_{j \in N} R_{i,j} x_i x_j - \sum_{i \in N} x_i L_i^0 \\
&= \sum_{i \in N} x_i \left(\sum_{j \in N} (R_{i,j} x_j) - L_i^0 \right),
\end{aligned}$$

where the first inequality comes from Proposition 2 and the final equation is equivalent to redefining Π using (24) with $\alpha = 0$.

As s_{int} was arbitrary, we have shown that for any solution to the integrated problem we can find an associated solution to the QWFP model given by (2)–(4) with $\alpha = 0$ in the definition of $\Pi_{i,i}$ in (24). Moreover, this solution to the QWFP has a higher objective value than the integrated problem. Hence, the QWFP model provides an upper bound to the integrated wind farm problem. $\square$

A.11 Proof: Proposition 8

Proof. Let s_{int} be an arbitrary solution to the integrated problem, with objective value z_{int} . Assume there are q turbines to place and that the distance to the substation exceeds δ for all but m turbines. Let z_{layout} and z_{cable} be the associated objective values for the WFLOP and WFCRP, respectively. Moreover, assume x is the solution vector of the WFLOP associated with solution s_{int} . Then,

$$\begin{aligned}
z_{int} &= z_{layout} - z_{cable} \\
&\leq z_{layout} - \sum_{i \in N} x_i d_i \frac{a_n}{c_n} - \left\lceil \frac{q-m}{c_n} \right\rceil \delta \left(a_1 - \frac{a_n}{c_n} \right) \\
&= \sum_{i \in N} \sum_{j \in N} R_{i,j} x_i x_j - \sum_{i \in N} x_i (\ell_i + 0 \cdot (u_i - \ell_i)) - \left\lceil \sum_{i \in N: d_i \geq \delta} \frac{x_i}{c_n} \right\rceil \delta \left(a_1 - \frac{a_n}{c_n} \right) \\
&\leq \sum_{i \in N} \sum_{j \in N} R_{i,j} x_i x_j - \sum_{i \in N} x_i (\ell_i + 0 \cdot (u_i - \ell_i)) - \sum_{i \in N: d_i \geq \delta} \frac{x_i}{c_n} \delta \left(a_1 - \frac{a_n}{c_n} \right) \\
&= \sum_{i \in N} \sum_{j \in N} R_{i,j} x_i x_j - \sum_{i \in N} x_i L_i^0 - \sum_{i \in N: d_i \geq \delta} x_i \frac{1}{c_n} \delta \left(a_1 - \frac{a_n}{c_n} \right) \\
&= \sum_{i \in N: d_i < \delta} x_i \left(\sum_{j \in N} (R_{i,j} x_j) - L_i^0 \right) + \sum_{i \in N: d_i \geq \delta} x_i \left(\sum_{j \in N} (R_{i,j} x_j) - L_i^0 - \frac{1}{c_n} \delta \left(a_1 - \frac{a_n}{c_n} \right) \right),
\end{aligned}$$

where the first inequality comes from Proposition 3 with $g = 1$ and the final equation is equivalent to redefining Π using (25) with $\alpha = 0$.

As s_{int} was arbitrary, we have shown that for any solution to the integrated problem we can find an associated solution to the QWFP model given by (2)–(4) with $\alpha = 0$ in the definition of $\Pi_{i,i}$ in (25). Moreover, this solution to the QWFP has a higher objective value than the integrated problem. Hence, the QWFP model provides an upper bound to the integrated wind farm problem. $\square$

A.12 Proof: Proposition 9

Proof. Let s_{int} be an arbitrary solution to the integrated problem, with objective value z_{int} . Assume there are q turbines to place and that the distance to the substation exceeds $\frac{1}{2}\delta$ for all but m turbines. Let z_{layout} and z_{cable} be the associated objective values for the WFLOP and WFCRP, respectively. Moreover, assume x is the solution vector of the WFLOP associated with solution

s_{int} . Then,

$$\begin{aligned}
z_{int} &= z_{layout} - z_{cable} \\
&\leq z_{layout} - \sum_{i \in N} x_i d_i \frac{a_n}{c_n} - (q - m) \ell_{min} \\
&= \sum_{i \in N} \sum_{j \in N} R_{i,j} x_i x_j - \sum_{i \in N} x_i (\ell_i + 0 \cdot (u_i - \ell_i)) - \sum_{i \in N: d_i \geq \frac{1}{2} \delta} x_i \ell_{min} \\
&= \sum_{i \in N} \sum_{j \in N} R_{i,j} x_i x_j - \sum_{i \in N} x_i L_i^0 - \sum_{i \in N: d_i \geq \frac{1}{2} \delta} x_i \ell_{min} \\
&= \sum_{i \in N: d_i < \frac{1}{2} \delta} x_i \left(\sum_{j \in N} (R_{i,j} x_j) - L_i^0 \right) + \sum_{i \in N: d_i \geq \frac{1}{2} \delta} x_i \left(\sum_{j \in N} (R_{i,j} x_j) - L_i^0 - \ell_{min} \right),
\end{aligned}$$

where the first inequality comes from Proposition 4 and the final equation is equivalent to redefining Π using (26) with $\alpha = 0$.

As s_{int} was arbitrary, we have shown that for any solution to the integrated problem we can find an associated solution to the QWFP model given by (2)–(4) with $\alpha = 0$ in the definition of $\Pi_{i,i}$ in (26). Moreover, this solution to the QWFP has a higher objective value than the integrated problem. Hence, the QWFP model provides an upper bound to the integrated wind farm problem. $\square$

B Appendix: Additional results

B.1 Results: WFLOP and WFCRP

Instance	Alpha α	WFLOP	WFCRP	Combined
A	no est	2024.933	21.87	2003.063
A	0	2024.832	22.65	2002.182
A	0.25	2024.234	20.03	2004.204
A	0.5	2024.234	20.03	2004.204
A	0.75	2023.254	19.26	2003.994
A	1	2022.796	19.30	2003.496
B	no est	4826.370	65.04	4761.330
B	0	4826.225	64.66	4761.565
B	0.25	4825.087	63.78	4761.307
B	0.5	4824.691	63.32	4761.371
B	0.75	4823.097	62.22	4760.877
B	1	4822.764	61.69	4761.074
C	no est	3987.677	47.04	3940.637
C	0	3987.101	45.84	3941.261
C	0.25	3986.535	45.20	3941.335
C	0.5	3985.686	44.47	3941.216
C	0.75	3984.616	44.07	3940.546
C	1	3983.433	42.69	3940.743
D	no est	8227.761	151.66	8076.101
D	0	8225.289	145.58	8079.709
D	0.25	8223.045	144.46	8078.585
D	0.5	8222.326	143.57	8078.756
D	0.75	8219.151	141.10	8078.051
D	1	8216.684	139.54	8077.144
E	no	6725.522	100.14	6625.382
E	0	6723.666	89.06	6634.606
E	0.25	6722.745	88.60	6634.145
E	0.5	6719.886	86.02	6633.866
E	0.75	6719.173	85.65	6633.523
E	1	6717.693	85.25	6632.443
F	no	7755.640	106.73	7648.910
F	0	7753.808	104.35	7649.458
F	0.25	7753.451	104.27	7649.181
F	0.5	7751.497	102.53	7648.967
F	0.75	7749.777	100.67	7649.107
F	1	7748.418	100.74	7647.678
G	no	6811.201	91.43	6719.771
G	0	6809.606	88.85	6720.756
G	0.25	6809.268	88.51	6720.758
G	0.5	6809.461	88.11	6721.351
G	0.75	6808.530	88.89	6719.640
G	1	6808.323	87.32	6721.003
H	no	9175.250	124.41	9050.840
H	0	9174.126	123.21	9050.916
H	0.25	9172.463	121.72	9050.743
H	0.5	9171.182	121.21	9049.972
H	0.75	9168.621	120.34	9048.281
H	1	9168.798	120.05	9048.748
I	no	14830.704	257.65	14573.054
I	0	14827.592	256.32	14571.272
I	0.25	14826.649	253.58	14573.069
I	0.5	14824.349	251.33	14573.019
I	0.75	14823.417	251.65	14571.767
I	1	14822.308	250.86	14571.448
J	no	11404.737	145.60	11259.137
J	0	11403.568	143.37	11260.198
J	0.25	11401.665	139.52	11262.145
J	0.5	11400.413	139.51	11260.903
J	0.75	11398.854	138.62	11260.234
J	1	11395.958	137.68	11258.278

Table 9: WFLOP, WFCRP, and Combined cost by instance and alpha value. Best found solutions are in boldface.

B.2 Results: effect of parameter α

α	WFLOP avg. from optimal	WFCRP avg. from max.	Cable est. avg. error	IWFLOP avg. from optimal
No est.	1.0000	0.9966	-	0.99968
0	0.9998	0.9737	0.2979	0.99986
0.25	0.9997	0.9529	0.0783	0.99996
0.5	0.9995	0.9441	0.2310	0.99994
0.75	0.9993	0.9341	0.4612	0.99983
1	0.9991	0.9263	0.7055	0.99978

Table 10: Effect of parameter α .

B.3 Results: cable bounds

Instance / zone	$\mathcal{U}$ (M€)	$\mathcal{L}_1$ (M€)	$\mathcal{L}_2$ (M€)	$\mathcal{L}_3$ (M€)	$\mathcal{L}_4$ (M€)	$\mathcal{L}_5$ (M€)
A / 0	12.76	7.42	8.00	8.29	4.24	8.29
A / 1	9.89	3.37	3.75	3.84	2.28	3.84
B / 0	64.66	45.93	47.66	49.22	15.99	49.22
C / 0	34.05	22.69	23.65	24.71	9.79	24.71
C / 1	11.79	5.74	6.33	6.76	4.90	6.76
D / 0	145.58	118.01	120.89	123.72	27.74	123.72
E / 0	1.82	0.79	0.99	0.99	0.98	0.99
E / 1	68.57	52.30	53.84	55.46	15.34	55.46
E / 2	18.67	11.20	11.78	12.41	5.88	12.41
F / 0	90.14	67.12	69.23	71.55	21.54	71.55
F / 1	14.21	8.75	9.33	9.63	4.24	9.63
G / 0	88.85	63.82	66.12	68.52	22.85	68.52
H / 0	106.61	81.51	84.20	86.82	25.79	86.82
H / 1	16.60	9.93	10.51	10.94	4.90	10.94
I / 0	256.32	200.81	205.80	211.29	50.92	211.29
J / 0	87.68	65.03	67.34	69.60	22.20	69.60
J / 1	44.32	31.16	32.51	33.65	12.08	33.65
J / 2	11.37	5.67	6.07	6.50	3.92	6.50

Table 11: For each zone the upper bound (solution from running MIP model) and initial lower bound is reported, alongside the lifted lower bounds, $\mathcal{L}_2$ to $\mathcal{L}_5$. Finally, we calculate the relative increase in the lower bound using $\mathcal{L}_5$. The relative increase is taken with respect to the upper bound. The results are from this paper.

Instance / zone	$\mathcal{U}$ (M€)	$\mathcal{L}_1$ (M€)	$\mathcal{L}_2$ (M€)	$\mathcal{L}_3$ (M€)	$\mathcal{L}_4$ (M€)	$\mathcal{L}_5$ (M€)
A / 0	10.05	3.64	5.08	4.32	6.71	6.71
A / 1	4.07	1.15	2.18	1.52	3.61	3.61
B / 0	55.61	31.80	37.95	34.62	25.28	37.95
C / 0	30.53	15.48	19.58	17.05	15.48	19.58
C / 1	12.38	4.85	6.90	5.63	7.74	7.74
D / 0	121.66	79.74	90.0	84.18	43.86	90.0
E / 0	1.06	0.22	0.22	0.38	0.00	0.38
E / 1	62.93	42.73	48.88	45.19	24.25	48.88
E / 2	10.01	3.94	5.99	4.88	9.29	9.29
F / 0	79.37	47.98	57.20	51.43	34.06	57.20
F / 1	10.46	4.00	6.05	4.68	6.71	6.71

Table 12: For each zone the upper bound (solution from running MIP model) and initial lower bound is reported, alongside the lifted lower bounds, $\mathcal{L}_2$ to $\mathcal{L}_5$. Finally, we calculate the relative increase in the lower bound using $\mathcal{L}_5$. The relative increase is taken with respect to the upper bound. The results are from the master thesis by Rustad [31] (with a branched topology), with the addition of $\mathcal{L}_\triangleright$, which was extended to a branched topology in this paper.